Review of Basic Concepts from Lecture 1

- Modeling and derivation of governing equation of motion for the free vibration of SDOF damped systems by Newton's law and energy method

$$m_{eq}\ddot{x}(t) + c_{eq}\dot{x}(t) + k_{eq}x(t) = 0$$

- Physics:

  - natural frequency $\omega_n = \sqrt{k_{eq}/m_{eq}}$

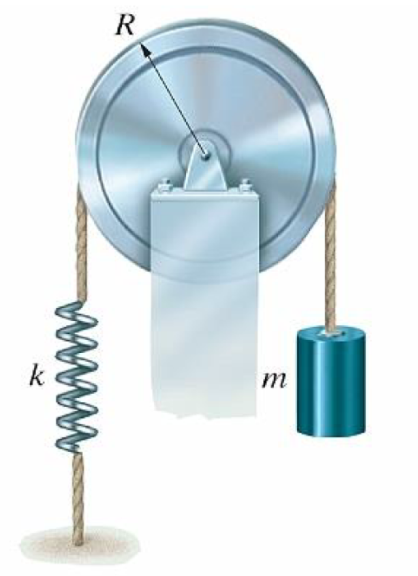
  - important concept of damping ratio $\zeta = \dfrac{c_{eq}}{2\sqrt{k_{eq}m_{eq}}}$

  - underdamped, critically damped, and overdamped motions as determined by the value of ζ

  - damped natural frequency: $\omega_d = \sqrt{1-\zeta^2}\,\omega_n$, if $\zeta < 1$

HW Problem:

The uniform pulley has radius R and mass moment of inertia I_O about its center of mass at O. Assume that the spring is linear, i.e., the force in the spring $F_s = k\Delta$, where Δ is the deflection of the spring from its unstretched configuration (natural length).



a) Draw the free-body diagram and derive the governing equation of motion for this system using an absolute coordinate x. The coordinate x measures the downward displacement of the mass relative to its position when the spring is unstretched. [Dynamics]
b) Suppose the mass is released from rest when the spring is unstretched, solve the governing differential equation analytically. (Remember to properly set the initial conditions.) [Differential equation]
c) Choose your own values of the parameters (justify your choice of the parametric values) or use the following suggested values to simulate the response of this system. Plot your analytical solution against the simulation result and compare. You may use MATLAB or any other suitable software. [Numerical simulation]

To simulate the response, we select the following values for the parameters. These values are selected based on an athlete using such a prototypical training machine in a gym. Note that the static deflection is about 2 cm.

$$m = 100 \text{ kg}, \quad k = 50{,}000 \text{ N/m}, \quad R = 0.15 \text{ m}, \quad I_O = 2 \text{ kg-m}^2$$

d) Write down the key observations of your solutions. What have you learned from this problem? [Observation of physics]

a) Draw the free-body diagram and derive the governing equation of motion for this system using an absolute coordinate x. The coordinate x measures the downward displacement of the mass relative to its position when the spring is unstretched.

We will derive the governing equation of motion based on the absolute coordinate and the relative coordinate measured from the equilibrium position, using both the Newton's law and the energy method.

Note: The Newton's law should only be applied to absolute coordinates. Hence, to do this problem correctly, one has to first derive the governing equation of motion using an absolute coordinate, and then rewrite the equation in terms of the relative coordinate.

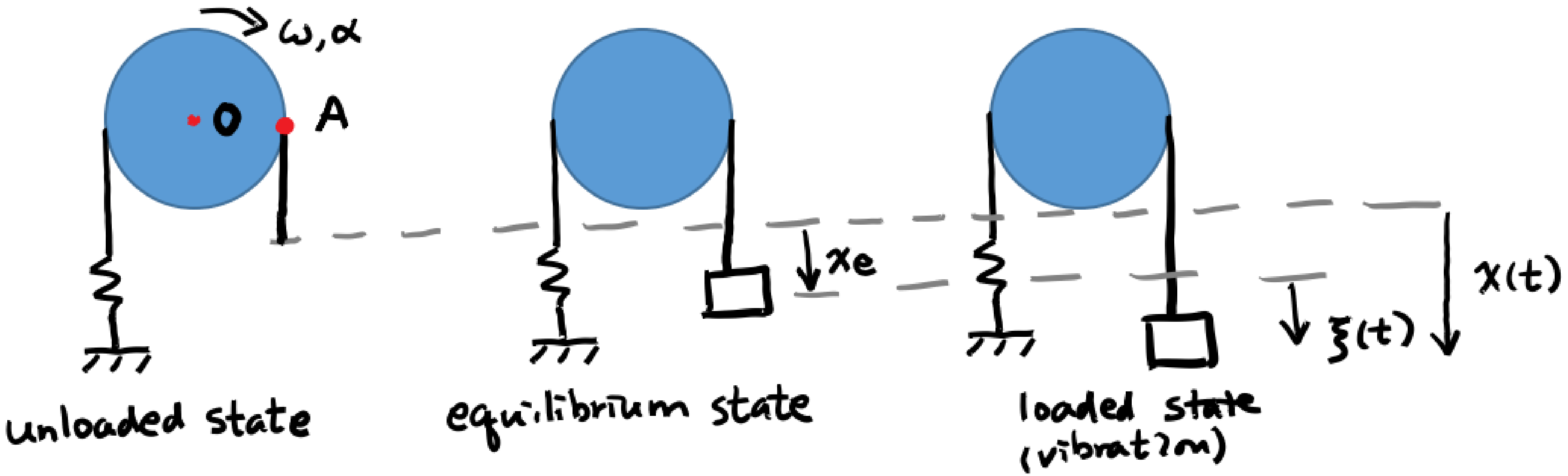
Note: For vibration systems in the vertical plane, it is convenient to derive the governing equation of motion using a coordinate measured relative from the equilibrium position to remove the contribution of the term(s) due to weight in the equation. The gravity basically just determines the equilibrium position. In general, within the context of linear vibration theory, vibration characteristics are independent of the static deflections.

Modeling and Critical Thinking:

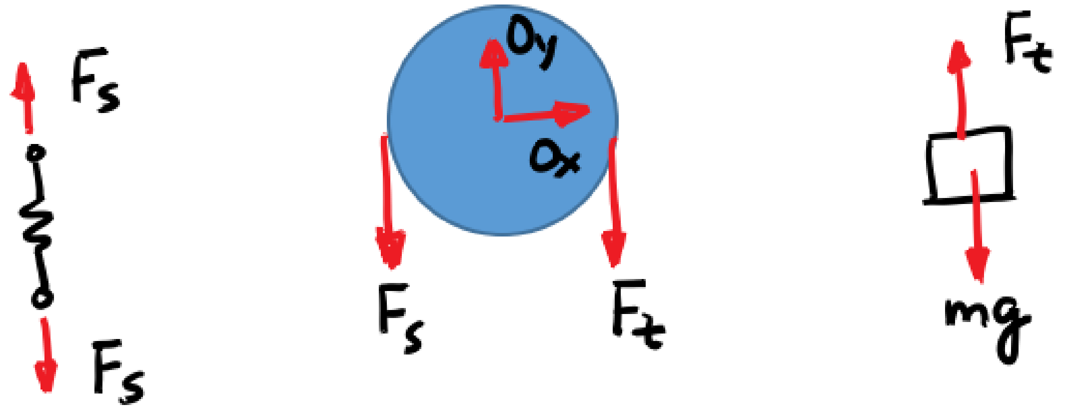
- Establish the coordinates. Let x be the absolute coordinate that measures the downward displacement of the mass relative to its position when the spring is unstretched. Let ξ be the coordinate relative to the equilibrium position.
- Assume the rope is taut, i.e., inextensible.
- Assume that the rope rolls without slip on the pulley. Thus, there must be friction on the pulley. Consider the point A in diagrams below. We have

$$\mathbf{v}_A = \mathbf{v}_0 + \boldsymbol{\omega} \times \mathbf{r}_{A/O} \;\Rightarrow\; -\dot{x}\,\mathbf{j} = 0 + (-\omega\,\mathbf{k}) \times R\,\mathbf{i} = -R\omega\,\mathbf{j}$$

  Hence, we have the kinematic relation $\dot{x} = R\omega$. Also, we have denoted ω and α as the positive, clockwise angular velocity and angular acceleration of the pulley, respectively.
- Assume the spring is massless. Hence, the tension forces on both ends of the spring are equal, as shown in the free-body diagram.



The free-body diagram is shown below.



Applying Newton's law to the mass gives:

$$mg - F_t = m\ddot{x} \tag{1}$$

The equation of angular motion for the pulley is:

$$I_O\alpha = (F_t - F_s)R \tag{2}$$

Invoking the kinematic relation $\ddot{x} = R\alpha$ and the spring kinetics model $F_s = kx$, and substituting these into the above equations give the governing equation of motion. Upon simplification, we obtain

$$I_O\frac{\ddot{x}}{R} = (mg - m\ddot{x} - kx)R \;\Rightarrow\; \overbrace{\left(m + \underbrace{\frac{I_O}{R^2}}_{\text{added inertia or mass}}\right)}^{m_{eq}}\ddot{x} + kx = mg \tag{3}$$

Now define the coordinate $\xi(t)$ relative to the equilibirum position x_e by the relation $\xi(t) = x(t) - x_e$. The equilibirum position is obtained by setting terms with time derivatives to zero. Hence, we have $kx_e = mg$, or $x_e = mg/k$. Substituting $x = x_e + \xi$ into Eq. (3) leads to:

$$\left(m + \frac{I_O}{R^2}\right)\ddot{\xi} + k\xi = 0 \tag{4}$$

Note that the equivalent mass for this system is $m_{eq} = m + I_O/R^2$. The natural frequency of vibration of this undamped system is:

$$\omega_n = \sqrt{\frac{k}{m + I_O/R^2}}$$

In your homework, derive the governing equation of motion in terms of the absolute coordinate $x(t)$ or the relative coordinate $\xi(t)$ by the energy principle. The difference in the form of the equations is just a simple coordinate transformation.

Using the energy method, we first establish relations for the kinetic energy and potential energy for this system. The kinetic energy is:

$$T = \frac{1}{2} m\dot{x}^2 + \frac{1}{2} I_O \omega^2 = \frac{1}{2} m_{eq} \dot{x}^2 = \frac{1}{2} m_{eq} \dot{\xi}^2 \tag{5}$$

The potential energy is due to the spring elasticity and the gravity. Set the datum of the gravity potential energy with respect to the unloaded state. Thus,

b) Suppose the mass is released from rest when the spring is unstretched, solve the governing differential equation analytically. (Remember to properly set the initial conditions.)

The mass is released when the spring is unstretched:

$$x(0) = x_0 = 0 \ \Rightarrow \ \xi(0) = \xi_0 = -x_e = -mg/k$$

The mass is released from rest: $\dot{x}(0) = v_0 = 0 \Rightarrow \dot{\xi}(0) = 0$

The general analytical solution of the vibration response in terms of $x(t)$ is:

$$x(t) = a\cos\omega_n t + b\sin\omega_n t + x_p(t)$$

3. Forced Response of SDOF Systems

- Derive the time and frequency response solutions of damped SDOF dynamic systems subjected to excitations

  Governing equation:

$$m_{eq}\ddot{x} + c_{eq}\dot{x} + k_{eq}x = F(t)$$

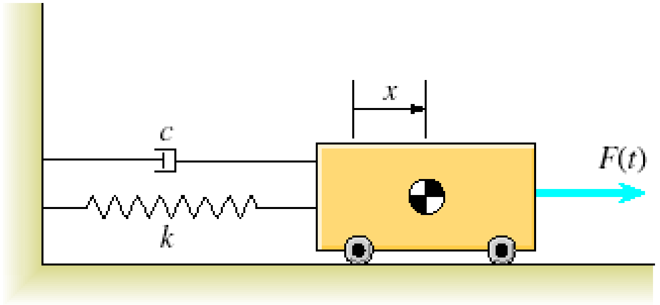
  Initial conditions:

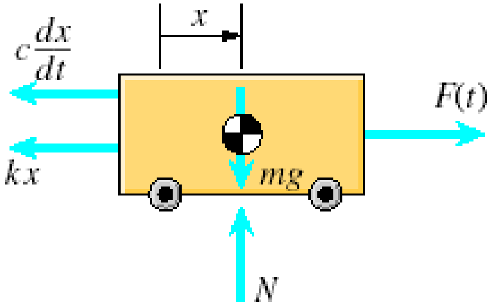
$$x(t = 0) = x_0, \qquad \dot{x}(0) = v_0$$

- The excitation $F(t)$ can be harmonic (one frequency), periodic (multiple frequencies) or non-perioidic

- Numerical solutions

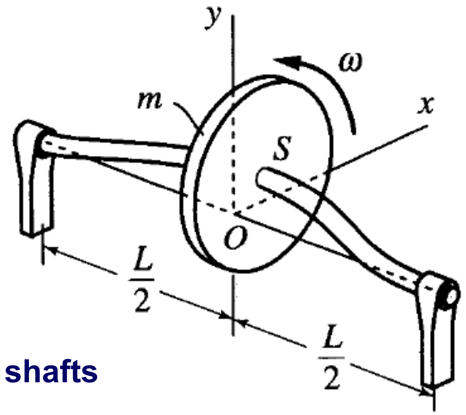
3.1 Forced Response to Periodic Excitations



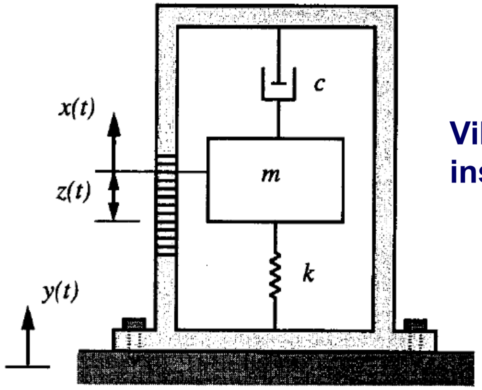


$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F(t) \quad \begin{cases} F(t) = F_0\cos(\omega t + \phi) & \textbf{Harmonic} \\ F(t) = \sum_{o}^{\infty}(a_k \cos k\omega t + b_k \sin k\omega t) & \textbf{Periodic (superposition principle)} \\ F(t) = g(t) & \textbf{Non-periodic} \end{cases}$$

Examples:



Whirling of rotating shafts



Vibration measuring instrument

3.1.1 Forced response to a harmonic excitation

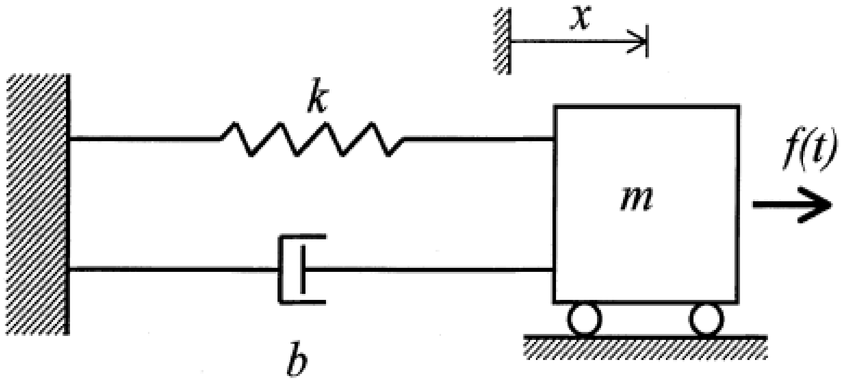


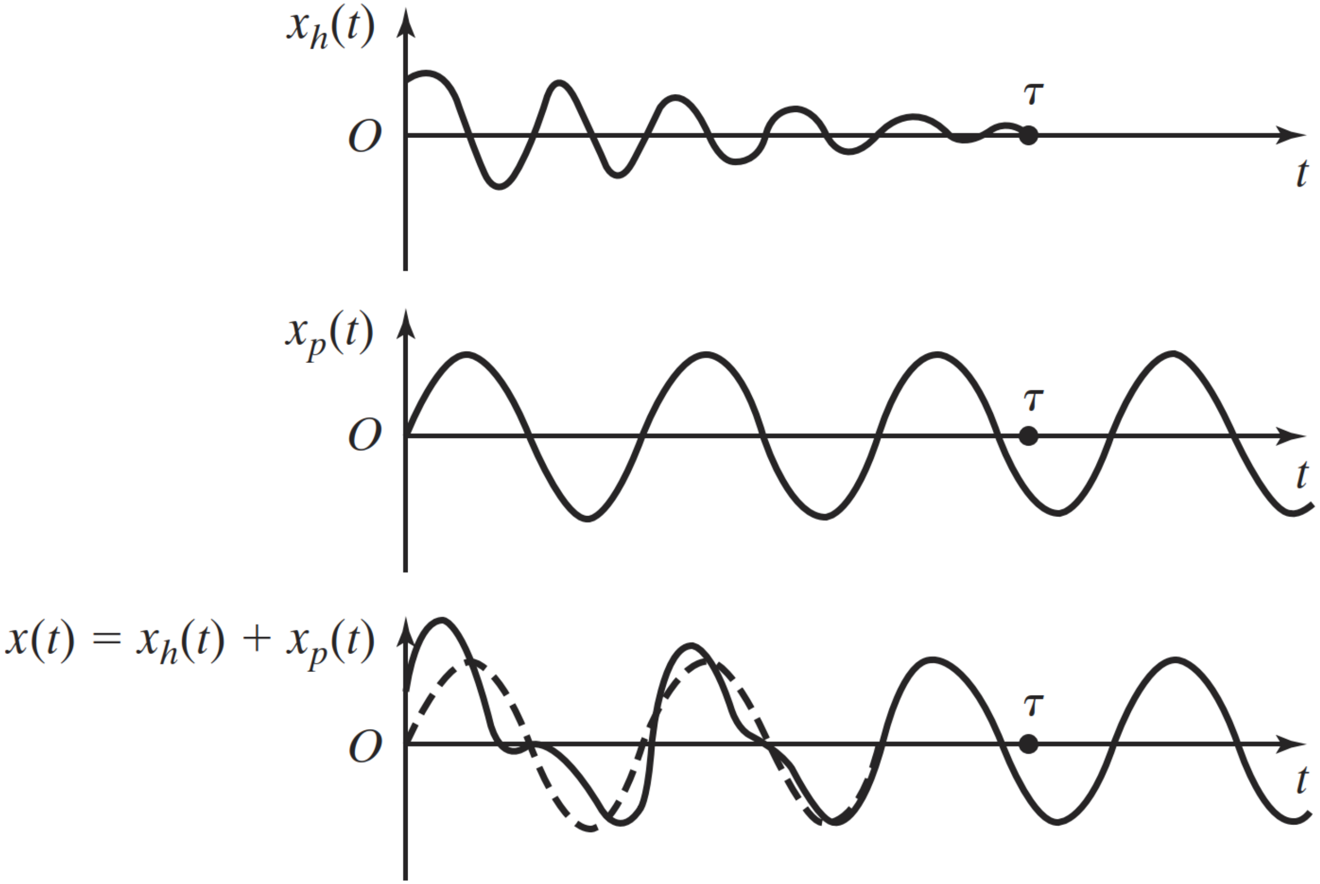
FIGURE 3.1 A forced simple oscillator.

$$\text{Total response} = \text{Homogenous solution } x_h(t) + \text{Particular solution } x_p(t)$$

Free response solution due to initial conditions. This response dissipates over time during the *transient* period.

+

Response to the excitation. Consider whether there might be resonance due to frequency matching. This response persists and is the *steady-state* response.



Source: Rao, *Mechanical Vibrations*, 6e.

Plots of homogeneous, particular, and total solutions of an **underdamped** example. As the homogenous solution decays over time, the particular solution becomes the *steady-state* response of the system.

Case 1: Undamped Oscillator with Excitation Frequency ≠ Natural Frequency

In this case,

$$\ddot{x} + \omega_n^2 x = a\cos\omega t \qquad \text{with} \qquad \omega \neq \omega_n \tag{3.8}$$

Homogeneous solution:

$$x_h = A_1\cos\omega_n t + A_2\sin\omega_n t \tag{3.9}$$

Particular solution:

$$x_p = \frac{a}{\left(\omega_n^2 - \omega^2\right)}\cos\omega t \tag{3.10}$$

$$\underbrace{x = A_1\cos\omega_n t + A_2\sin\omega_n t}_{H} \quad + \quad \underbrace{\frac{a}{\left(\omega_n^2 - \omega^2\right)}\cos\omega t}_{P} \tag{3.11}$$

Satisfies the homogeneous equation — Satisfies the equation with input

Substituting the initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$ into $x(t)$ gives the constant A_1 and A_2.

$$x_o = A_1 + \frac{a}{\omega_n^2 - \omega^2} \tag{3.13a}$$

$$v_o = A_2\omega_n \tag{3.13b}$$

Hence, the complete response is

$$\underbrace{x = \left[x_o - \frac{a}{(\omega_n^2 - \omega^2)}\right]\cos\omega_n t + \frac{v_o}{\omega_n}\sin\omega_n t}_{H} \quad + \quad \underbrace{\frac{a}{(\omega_n^2 - \omega^2)}\cos\omega t}_{P} \tag{3.14a}$$

Homogeneous solution — Particular solution

$$= \underbrace{x_o\cos\omega_n t + \frac{v_o}{\omega_n}\sin\omega_n t}_{X} \quad + \quad \underbrace{\frac{a}{(\omega_n^2 - \omega^2)}\underbrace{\left[\cos\omega t - \cos\omega_n t\right]}_{2\sin\frac{(\omega_n+\omega)}{2}t\ \sin\frac{(\omega_n-\omega)}{2}t}}_{F}$$

Free response
(depends only on ICs)
Comes from x_h.
*Sinusodal at ω_n.

Forced response (depends on input)
Comes from both x_h and x_p.
*Will exhibit a beat phenomenon for small $\omega_n - \omega$; i.e., $\frac{(\omega_n + \omega)}{2}$ wave *modulated* by $\frac{(\omega_n - \omega)}{2}$ wave.

This is a stable response in the sense of bounded-input-bounded-output (BIBO) stability, as it is bounded and does not increase steadily.

Case 2: Undamped Oscillator with $\omega = \omega_n$ (Resonant condition)

In this case, the x_p that was used before is no longer valid (this is the degenerate case), because otherwise the particular solution cannot be distinguished from the homogeneous solution and the former will be completely absorbed into the latter. Instead, in view of the double-integration nature of the forced system equation when $\omega = \omega_n$, use the particular solution (P):

$$x_p = \frac{at}{2\omega}\sin\omega t \tag{3.15}$$

This choice of particular solution is strictly justified by the fact that it satisfies the forced system equation.

Complete solution:

$$x = A_1\cos\omega t + A_2\sin\omega t + \frac{at}{2\omega}\sin\omega t \tag{3.16}$$

Substituting the initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$ into $x(t)$ gives the constant A_1 and A_2.

$$\underbrace{x = x_o\cos\omega t + \frac{v_o}{\omega}\sin\omega t}_{X=H} \qquad + \qquad \underbrace{\frac{at}{2\omega}\sin\omega t}_{F=P} \tag{3.18}$$

Free response (depends on ICs)
*Sinusodal at ω

Forced response (depends on input)
*Amplitude increases linearly

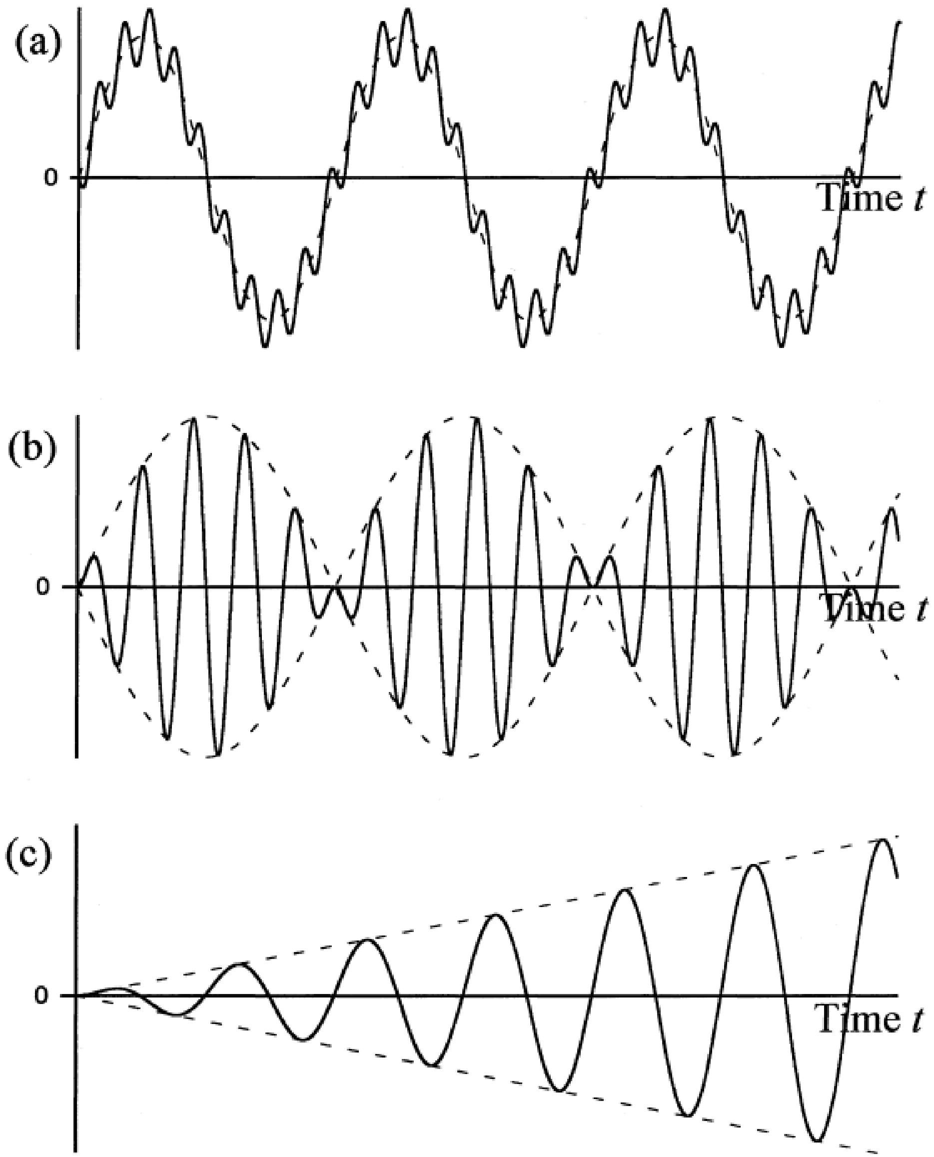


FIGURE 3.2 Forced response of a harmonically excited undamped simple oscillator: (a) for a large frequency difference, (b) for a small frequency difference (beat phenomenon), and (c) response at resonance.

$\omega \neq \omega_n$

$\omega \sim \omega_n$
beating

$\omega = \omega_n$
resonance

Excessive vibration leads to excessive stress and possibly structural failure

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Case 3: Damped Oscillator

The equation of forced motion is

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = a\cos\omega t \qquad (3.19)$$

Particular Solution (Method 1)

Since derivatives of both odd order and even order are present in this equation, the particular solution should have terms corresponding to odd and even derivatives of the forcing function (i.e., sin ωt and cos ωt). Hence, the appropriate particular solution will be of the form

$$x_p = a_1\cos\omega t + a_2\sin\omega t \qquad (3.20)$$

Substitute equation (3.20) in (3.19) to obtain

$$-\omega^2 a_1\cos\omega t - \omega^2 a_2\sin\omega t + 2\zeta\omega_n\left[-\omega a_1\sin\omega t + \omega a_2\cos\omega t\right] +$$

$$\omega_n^2\left[a_1\cos\omega t + a_2\sin\omega t\right] = a\cos\omega t$$

Solving the above two equations for the constant A_1 and A_2.

This is called the method of undetermined coefficients. The solution is:

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{D} \begin{bmatrix} (\omega_n^2 - \omega^2) & -2\zeta\omega_n\omega \\ 2\zeta\omega_n\omega & (\omega_n^2 - \omega^2) \end{bmatrix} \begin{bmatrix} a \\ 0 \end{bmatrix} \tag{3.22}$$

with the determinant

$$D = (\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2 \tag{3.23}$$

On simplification,

$$a_1 = \frac{(\omega_n^2 - \omega^2)}{D} a \tag{3.24a}$$

$$a_2 = \frac{2\zeta\omega_n\omega}{D} a \tag{3.24b}$$

The homogeneous solution is the same as the free vibration response, depending on whether the system is underdamped, critically damped, or overdamped. Again, apply the initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$ to obtain the constants in the homogenous solution.

Particular Solution (Method 2): Complex Function Method

Consider

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = ae^{j\omega t} \tag{3.25}$$

where the excitation is complex. (*Note*: $e^{j\omega t} = \cos\omega t + j\sin\omega t$).
The resulting "complex" particular solution is

$$x_p = X(j\omega)e^{j\omega t}. \tag{3.26}$$

Note that one should take the "real part" of this solution as the true particular solution.

First substitute equation (3.26) into (3.25):

$$X\left[-\omega^2 + 2\zeta\omega_n j\omega + \omega_n^2\right]e^{j\omega t} = ae^{j\omega t}$$

Hence (since $e^{j\omega t} \neq 0$ in general),

$$X = \frac{a}{\left[-\omega^2 + 2\zeta\omega_n j\omega + \omega_n^2\right]} \tag{3.27}$$

Note that setting the denominator of Eq. (3.27) to zero gives the frequency equation.

Response Magnification Factor under Harmonic Excitation

Consider the governing equation of the form:

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$

The steady-state response expressed in terms of the magnification factor:

$$X = \frac{F_0}{[(k - m\omega^2)^2 + c^2\omega^2]^{1/2}}\,, \qquad \phi = \tan^{-1}\left(\frac{c\omega}{k - m\omega^2}\right)$$

Define static deflection $\delta_{st} = F_0/k$, and frequency ratio $r = \omega/\omega_n$

$$\frac{X}{\delta_{st}} = \underbrace{\frac{1}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}}}_{\text{magnification factor } M} = |H(i\omega)|\,, \qquad \phi = \tan\left(\frac{2\zeta r}{1 - r^2}\right)$$

Note that the response is in term of the damping ratio and frequency ratio only. Here, $H(i\omega)$ is the complex frequency response function or *transfer function*. In some books, the transfer function is denoted as $G(i\omega)$ instead of $H(i\omega)$.

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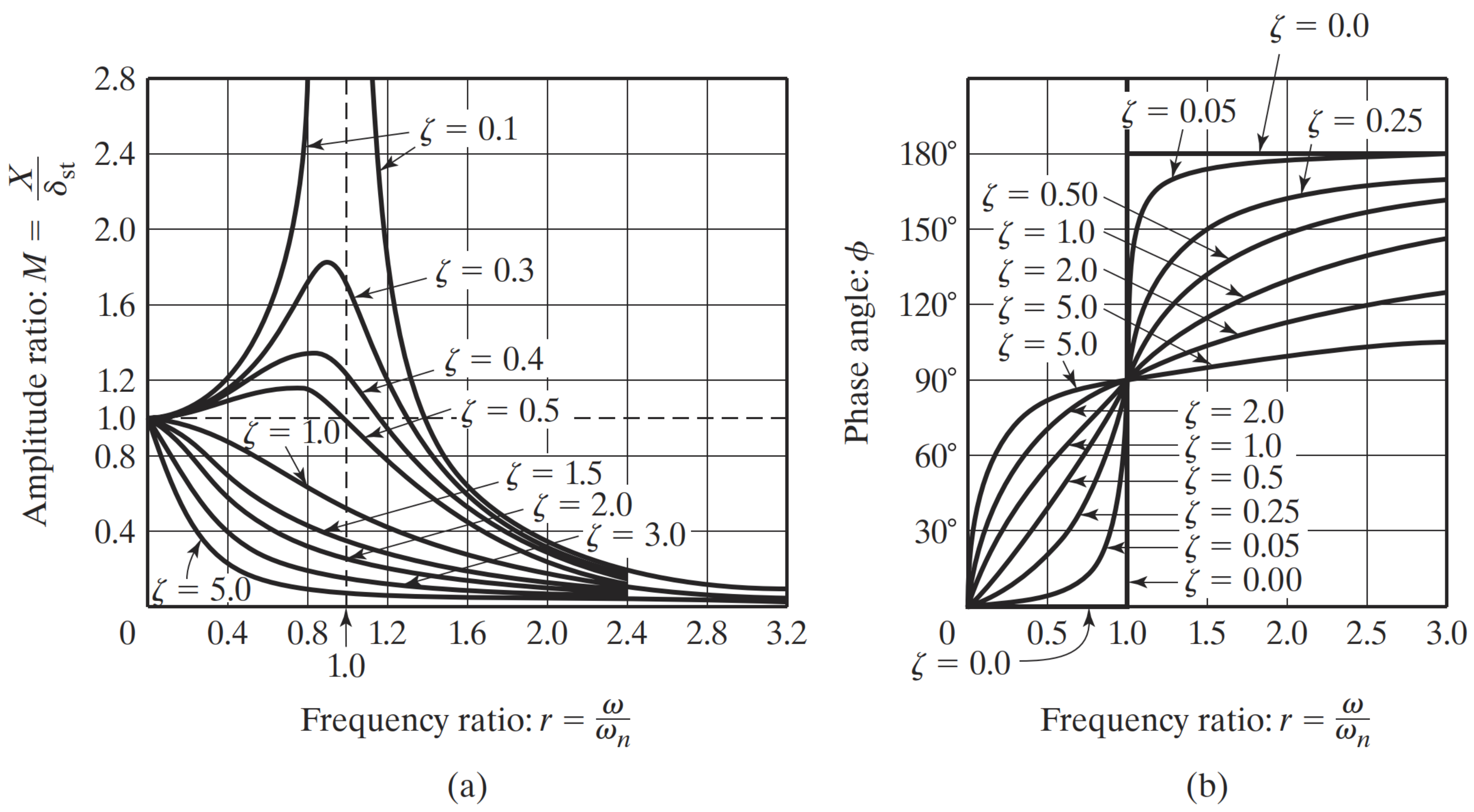

FIGURE 3.11 Variation of X and ϕ with frequency ratio r.

Source: Rao, *Mechanical Vibrations*, 6e.

Example:

Find the total response of a single-degree-of-freedom system with $m = 10$ kg, $c = 20$ N-s/m, $k = 4000$ N/m, $x_0 = 0.01$ m, and $\dot{x}_0 = 0$ under the following conditions:

a. An external force $F(t) = F_0 \cos \omega t$ acts on the system with $F_0 = 100$ N and $\omega = 10$ rad/s.
b. Free vibration with $F(t) = 0$.

To solve for the response, we first determine the key parameters (undamped natural frequency, damping ratio, damped natural frequency, etc.) in order to understand what formulas to apply.

Source: Rao, *Mechanical Vibrations*, 6e.

Solution:

a. From the given data, we obtain

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{4000}{10}} = 20 \text{ rad/s}$$

$$\delta_{st} = \frac{F_0}{k} = \frac{100}{4000} = 0.025 \text{ m}$$

$$\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{km}} = \frac{20}{2\sqrt{(4000)(10)}} = 0.05$$

underdamped

$$\omega_d = \sqrt{1-\zeta^2}\omega_n = \sqrt{1-(0.05)^2}(20) = 19.974984 \text{ rad/s}$$

$$r = \frac{\omega}{\omega_n} = \frac{10}{20} = 0.5$$

$$X = \frac{\delta_{st}}{\sqrt{(1-r^2)^2 + (2\,\zeta\, r)^2}} = \frac{0.025}{\left[(1-0.05^2)^2 + (2\cdot 0.5\cdot 0.5)^2\right]^{1/2}} = 0.03326 \text{ m} \qquad \text{(E.1)}$$

$$\phi = \tan^{-1}\left(\frac{2\zeta r}{1-r^2}\right) = \tan^{-1}\left(\frac{2\cdot 0.05\cdot 0.5}{1-0.5^2}\right) = 3.814075^\circ \qquad \text{(E.2)}$$

Amplitude and phase of the forced steady-state response.

Solution (cont.):

The complete solution is given by $x(t) = x_h(t) + x_p(t)$, where $x_h(t)$ is given by Eq. (2.70 f). Thus, for an underdamped system, we have

$$x(t) = X_0 e^{-\zeta\omega_n t}\cos(\omega_d t - \phi_0) + X\cos(\omega t - \phi) \tag{3.35}$$

$$\omega_d = \sqrt{1 - \zeta^2}\,\omega_n$$

where X and ϕ are given by Eqs. (3.30) and (3.31), respectively, and X_0 and ϕ_0 [denoted as X and ϕ in Eq. (2.70f)] can be determined from the initial conditions. For the initial conditions, $x(t = 0) = x_0$ and $\dot{x}(t = 0) = \dot{x}_0$, Eq. (3.35) yields

$$\begin{aligned} x_0 &= X_0 \cos\phi_0 + X\cos\phi \\ \dot{x}_0 &= -\zeta\omega_n X_0 \cos\phi_0 + \omega_d X_0 \sin\phi_0 + \omega X \sin\phi \end{aligned} \tag{3.36}$$

The solution of Eq. (3.36) gives X_0 and ϕ_0 as

$$\left.\begin{aligned} X_0 &= \left[(x_0 - X\cos\phi)^2 + \frac{1}{\omega_d^2}(\zeta\omega_n x_0 + \dot{x}_0 - \zeta\omega_n X\cos\phi - \omega X\sin\phi)^2\right]^{\frac{1}{2}} \\ \tan\phi_0 &= \frac{\zeta\omega_n x_0 + \dot{x}_0 - \zeta\omega_n X\cos\phi - \omega X\sin\phi}{\omega_d(x_0 - X\cos\phi)} \end{aligned}\right\} \tag{3.37}$$

Solution (cont.):

Recall:

$$x_0 = X_0 \cos \phi_0 + X \cos \phi$$
$$\dot{x}_0 = -\zeta \omega_n X_0 \cos \phi_0 + \omega_d X_0 \sin \phi_0 + \omega X \sin \phi \qquad (3.36)$$

Using the initial conditions, $x_0 = 0.01$ and $\dot{x}_0 = 0$, Eq. (3.36) yields:

$$0.01 = X_0 \cos \phi_0 + (0.03326)(0.997785)$$

or

$$X_0 \cos \phi_0 = -0.023186 \qquad \text{(E.3)}$$

$$0 = -(0.05)(20) X_0 \cos \phi_0 + X_0 (19.974984) \sin \phi_0 + (0.03326)(10) \sin (3.814075^\circ) \qquad \text{(E.4)}$$

Substituting the value of $X_0 \cos \phi_0$ from Eq. (E.3) into (E.4), we obtain

$$X_0 \sin \phi_0 = -0.002268 \qquad \text{(E.5)}$$

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Solution (cont.):

Solution of Eqs. (E.3) and (E.5) yields

$$X_0 = \left[(X_0 \cos \phi_0)^2 + (X_0 \sin \phi_0)^2 \right]^{1/2} = 0.023297 \tag{E.6}$$

and

$$\tan \phi_0 = \frac{X_0 \sin \phi_0}{X_0 \cos \phi_0} = 0.0978176$$

or

$$\phi_0 = 5.586765^\circ \tag{E.7}$$

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Solution (cont.):

b. For free vibration, the total response is given by

$$x(t) = X_0 e^{-\zeta\,\omega_n\,t} \cos\left(\omega_d t - \phi_0\right) \tag{E.8}$$

Using the initial conditions $x(0) = x_0 = 0.01$ and $\dot{x}(0) = \dot{x}_0 = 0$, X_0 and ϕ_0 of Eq. (E.8) can be determined as (see Eqs. (2.73) and (2.75)):

$$X_0 = \left[x_0^2 + \left(\frac{\zeta\omega_n x_0}{\omega_d}\right)^2\right]^{1/2} = \left[0.01^2 + \left(\frac{0.05 \cdot 20 \cdot 0.01}{19.974984}\right)^2\right]^{1/2} = 0.010012 \tag{E.9}$$

$$\phi_0 = \tan^{-1}\left(\frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d\, x_0}\right) = \tan^{-1}\left(\frac{0.05 \cdot 20}{19.974984}\right) = 2.865984^\circ \tag{E.10}$$

Note that the constants X_0 and ϕ_0 in cases (a) and (b) are very different.

Note:

$$X = X_0 = \sqrt{(C_1')^2 + (C_2')^2} = \frac{\sqrt{x_0^2\omega_n^2 + \dot{x}_0^2 + 2x_0\dot{x}_0\zeta\omega_n}}{\sqrt{1-\zeta^2}\,\omega_n} \tag{2.73}$$

$$\phi = \tan^{-1}\left(\frac{C_2'}{C_1'}\right) = \tan^{-1}\left(\frac{\dot{x}_0 + \zeta\omega_n x_0}{x_0\omega_n\sqrt{1-\zeta^2}}\right) \tag{2.75}$$

Note that, for underdamped systems, we have used the following expression for the homogeneous solution:

$$x(t) = X\mathrm{e}^{-\zeta\omega_n t}\cos(\omega_d t - \phi)$$

where the amplitude X and the phase ϕ <u>for free response</u> are determined by invoking the initial conditions $x(0) = x_0$ and $\dot{x}(0) = v_0$.

$$X = \frac{1}{\omega_d}\sqrt{(x_0\omega_d)^2 + (v_0 + \zeta\omega_n x_0)^2}\,, \qquad \phi = \tan^{-1}\left(\frac{v_0 + \zeta\omega_\mathrm{n} x_0}{x_0\omega_d}\right)$$

An alternative solution form (maybe easier) can also be given by:

$$x(t) = e^{-\zeta\omega_n t}\left(x_0\cos\omega_d t + \frac{v_0 + \zeta\omega_n x_0}{\omega_d}\sin\omega_d t\right), \qquad \text{for } \zeta \neq 1$$

$$x(t) = x_0 e^{-\omega_n t} + (v_0 + \omega_n x_0)te^{-\omega_n t}\,, \qquad \text{for } \zeta = 1$$

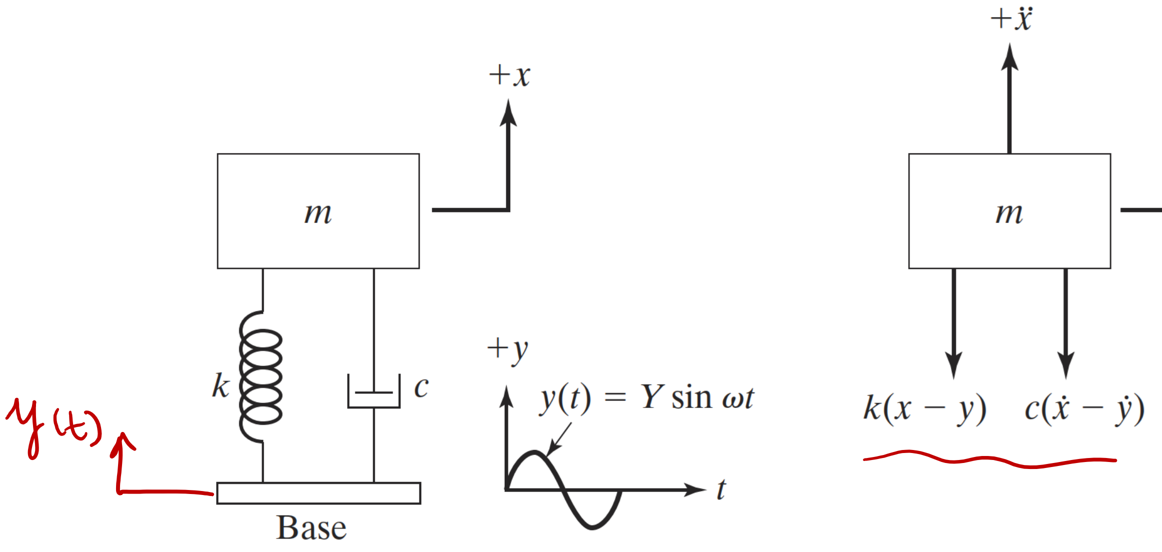
Response under base excitation

The governing equation of motion is:

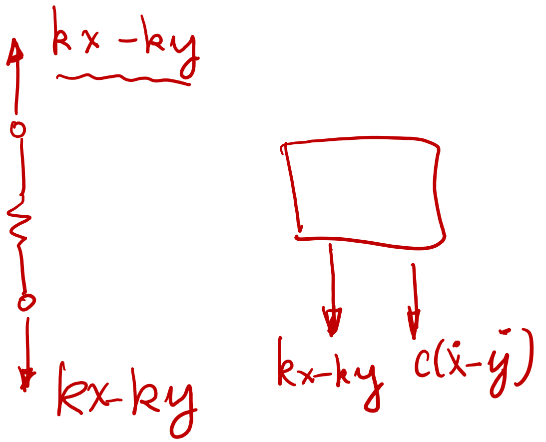
$$m\ddot{x} = -k(x-y) - c(\dot{x}-\dot{y}) \Rightarrow \; m\ddot{x} + c\dot{x} + kx = ky + c\dot{y}$$

If the base excitation $y(t) = Y\sin\omega t$, then the governing equation becomes:

$$m\ddot{x} + c\dot{x} + kx = kY\sin\omega t + c\omega Y\cos\omega t$$



(a) (b)



Source: Rao, *Mechanical Vibrations*, 6e.

Response under base excitation

Let the steady-state response (particular solution) be expressed as

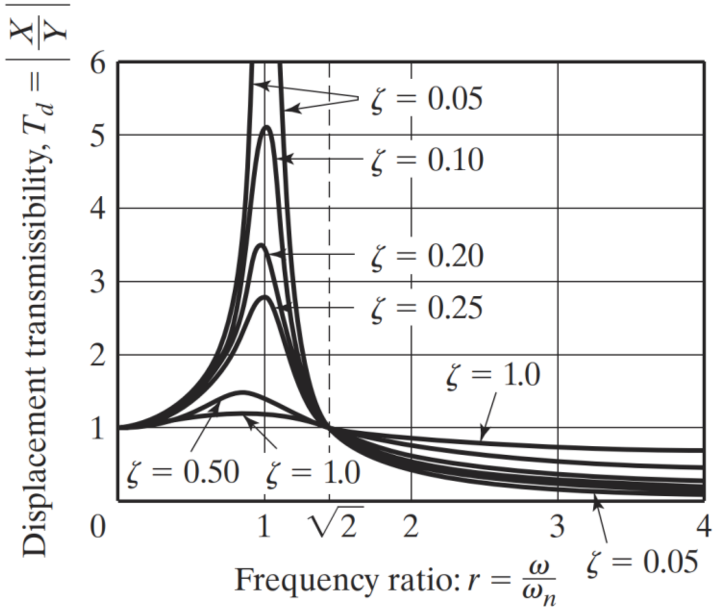
$$x_p(t) = X\sin(\omega t - \phi)$$

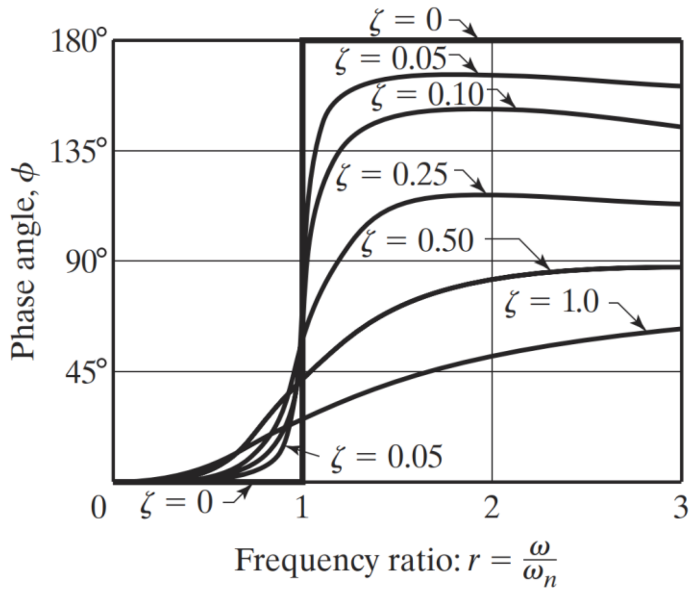
The *displacement transmissibility* is:

$$\frac{X}{Y} = \left[\frac{1+(2\zeta r)^2}{(1-r^2)^2+(2\zeta r)^2}\right]^{\frac{1}{2}} = \sqrt{1+(2\zeta r)^2}\,|H(i\omega)|, \qquad \phi = \tan\left[\frac{2\zeta r^3}{1+(4\zeta^2-1)r^2}\right]$$

The displacement transmissibility T_d attains a maximum for $0 < \zeta < 1$ at frequency $r_m < 1$:

$$r_m = \frac{1}{2\zeta}\left[\sqrt{1+8\zeta^2}-1\right]^{1/2}$$





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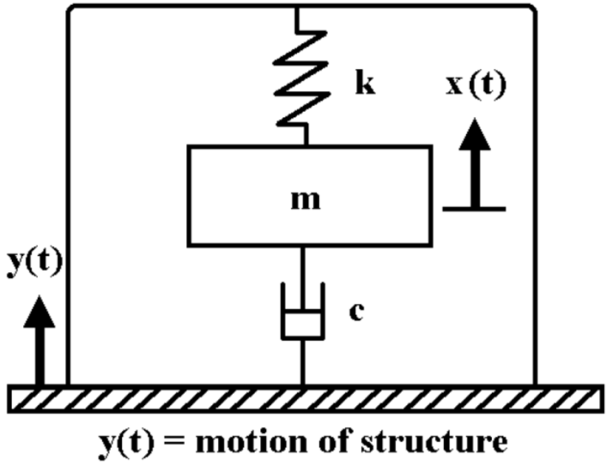
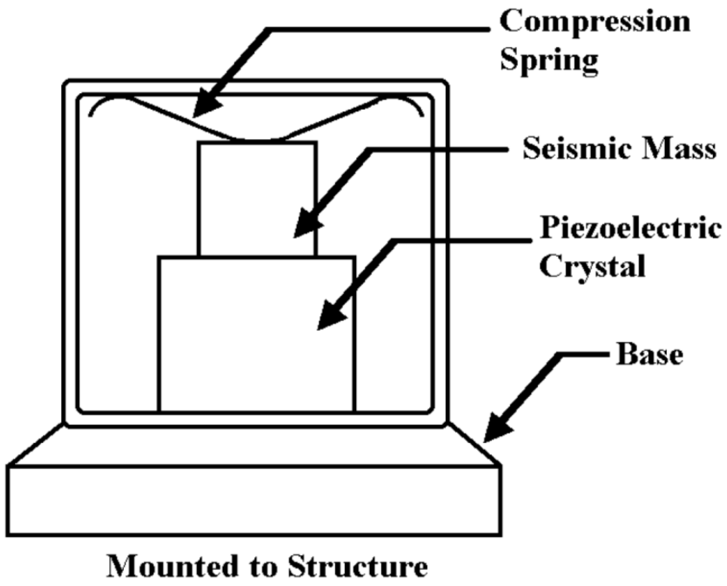
Response under base excitation – Accelerometer:

Steady-state response to base excitations can be applied to understand the operations of accelerometers shown. Accelerometers measure the relative displacement: $z(t) = x(t) - y(t)$. The governing equation of motion becomes

$$m\ddot{z} + c\dot{z} + kz = -m\ddot{y} = m\omega^2 Y \sin \omega t$$





Response under base excitation – Accelerometer:

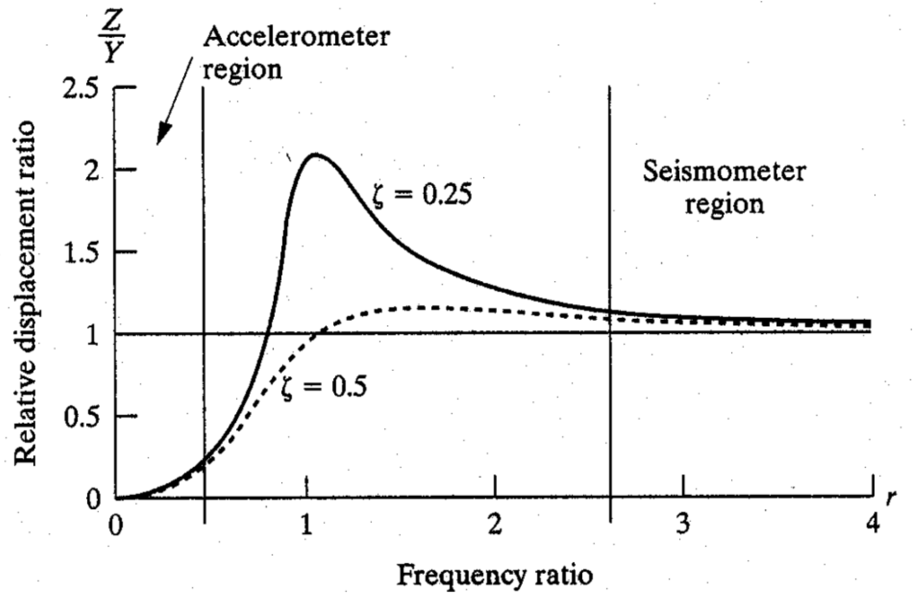
Let the steady-state response (particular solution) be expressed as

$$z_p(t) = Z\sin(\omega t - \phi)$$

The *displacement transmissibility* is:

$$\frac{Z}{Y} = r^2\left[\frac{1}{(1-r^2)^2 + (2\zeta r)^2}\right]^{1/2} = r^2\,|H(i\omega)|, \qquad \phi = \tan\left[\frac{2\zeta r}{(1-r^2)}\right]$$

In applications, for large values of $r = \omega/\omega_n$ (typically, $r > 3$), $Z \cong Y$. Hence, the relative displacement of the accelerometer can be used to estimate the amplitude of the base excitation, such as seismometers for applications when the excitation frequency is larger than the natural frequency.



Response under base excitation – Accelerometer:

Since $r^2 = \omega^2/\omega_n^2$, rewrite the solution in terms of the acceleration

$$\omega_n^2\, z(t) = |H(i\omega)|\, \omega^2 Y \sin(\omega t - \phi) \Rightarrow -\omega_n^2\, z(t) = |H(i\omega)|\, \ddot{y}(t)$$

$$\text{Thus,} \qquad \frac{\text{measured acceleration}}{\text{true acceleration}} = |H(i\omega)| = \left[\frac{1}{(1-r^2)^2 + (2\zeta r)^2}\right]^{1/2}$$

$z(t) = Z e^{i\omega t}$

$|\ddot{z}| = \omega^2 Z e^{i\omega t}$

For small values of r, $|H(i\omega)| \approx 1$, so $\Rightarrow -\omega_n^2\, z(t) \approx \ddot{y}(t)$.

Accelerometers are used for small r values to measure the acceleration of a base excitation or a vibrating body. Applications include mass resting on a piezoelectric ceramic crystal (Kistler 8694, high-frequency model with $\omega_n \cong 80$ kHz). Up to excitations of 16 kHz can be measured ($0 < r < 0.2$).

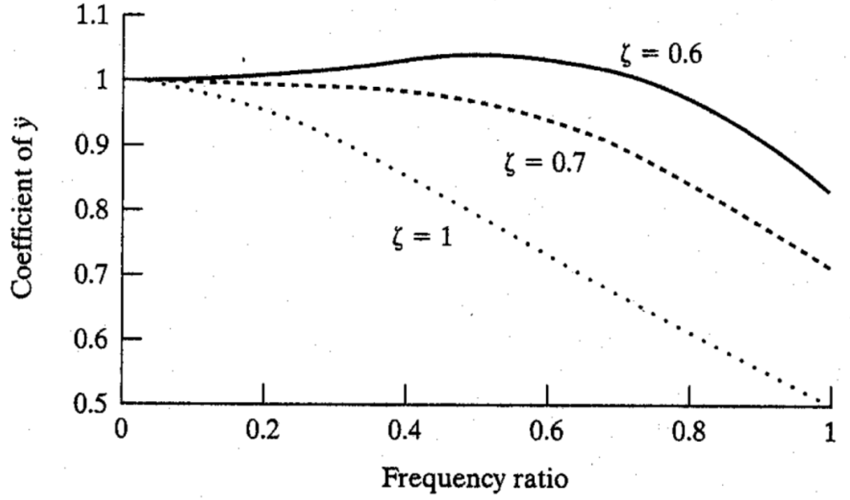


Figure 2.24 Effect of damping on the constant of proportionality between base acceleration and the relative displacement (voltage) for an accelerometer.

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Rao, S.S., *Mechanical Vibrations*, 6e, Example 10.3

Design of an Accelerometer

An accelerometer has a suspended mass of 0.01 kg with a damped natural frequency of vibration of 150 Hz. When mounted on an engine undergoing an acceleration of $1g$ at an operating speed of 6000 rpm, the acceleration is recorded as 9.5 m/s^2 by the instrument. Find the damping constant and the spring stiffness of the accelerometer.

The key to solving this problem is to find the relation between the frequency ratio r and the damping ratio ζ. Given that $m = 0.01$ kg, $\omega_d = 150$ Hz $= 942.48$ rad/s, $\omega = 6000$ rpm $= 628.32$ rad/s. From these given data, we can obtain a relation of r in terms of ζ

$$\frac{\omega}{\omega_d} = \frac{\omega}{\omega_n\sqrt{1-\zeta^2}} = \frac{r}{\sqrt{1-\zeta^2}} \Rightarrow \; r = 0.667\sqrt{1-\zeta^2} \Rightarrow r^2 = 0.444\left(1-\zeta^2\right) \qquad (1)$$

Using the relation,

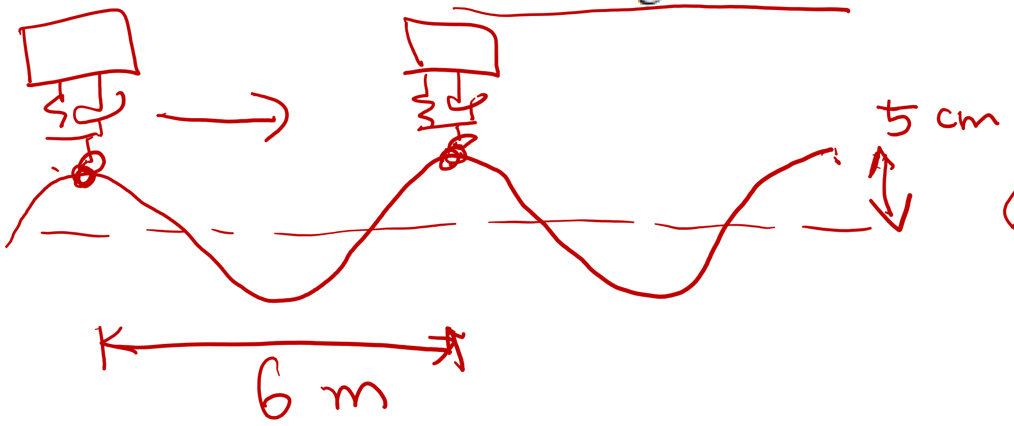
$$\frac{\text{measured acc.}}{\text{true acc.}} = \frac{9.5}{9.81} = |H(i\omega)| = \left[\frac{1}{(1-r^2)^2+(2\zeta r)^2}\right]^{1/2} \Rightarrow \left(1-r^2\right)^2 + (2\zeta r)^2 = 1.066 \qquad (2)$$

Solving the above two equations in two unknowns to obtain the answers. Alternatively, if one eliminates r from these two equations, a quadratic equation in ζ will be resulted, and a root finding scheme, such as the standard formula or MATLAB `roots`, `fzero`, etc. needs to be applied to solve for the answer for ζ.

Example:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{400{,}000}{1200}} \simeq 18.25 \text{ rad/s}$$

Figure 3.18 shows a simple model of a motor vehicle that can vibrate in the vertical direction while traveling over a rough road. The vehicle has a mass of 1200 kg. The suspension system has a spring constant of 400 kN/m and a damping ratio of $\zeta = 0.5$. If the vehicle speed is 20 km/h, determine the displacement amplitude of the vehicle. The road surface varies sinusoidally with an amplitude of $Y = 0.05$ m and a wavelength of 6 m.



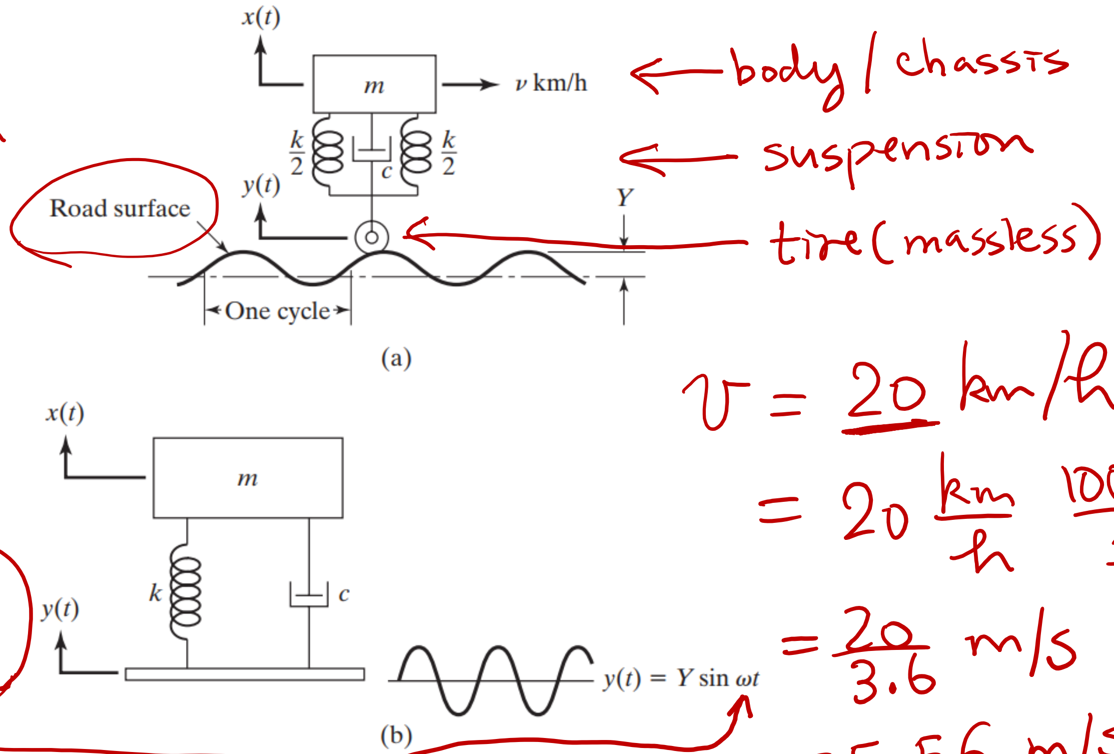


FIGURE 3.18 Vehicle moving over a rough road.

$$v = 20 \text{ km/h}$$

$$= 20\,\frac{\text{km}}{\text{h}}\;\frac{1000\text{m}}{1\text{km}}\;\frac{1\text{h}}{3600\text{sec}}$$

$$= \frac{20}{3.6} \text{ m/s}$$

$$= 5.56 \text{ m/s}$$

$$\text{time} = \frac{6}{5.56} = 1.079 \text{ sec}$$

period

$$T = \frac{2\pi}{\omega} \Rightarrow \omega = \frac{2\pi}{T}$$

$$T = \frac{\lambda}{v}$$

$$\omega = 2\pi v/\lambda$$

Solution:

Solution: The frequency ω of the base excitation can be found by dividing the vehicle speed v km/hr by the length of one cycle of road roughness:

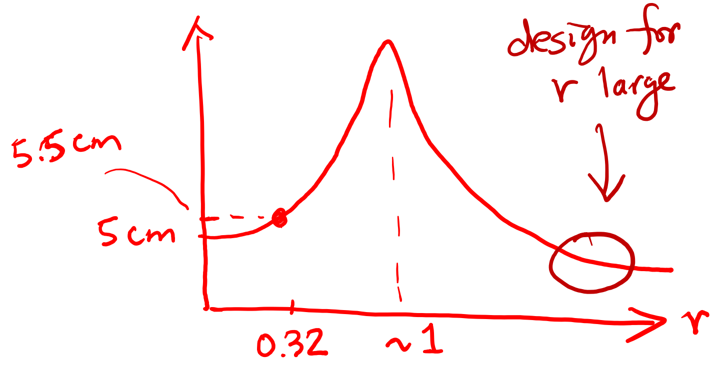
$$\omega = 2\pi f = 2\pi\left(\frac{v \times 1000}{3600}\right)\frac{1}{6} = 0.290889v \text{ rad/s}$$

For $v = 20$ km/hr, $\omega = 5.81778$ rad/s. The natural frequency of the vehicle is given by

$$\omega_n = \sqrt{\frac{k}{m}} = \left(\frac{400 \times 10^3}{1200}\right)^{1/2} = 18.2574 \text{ rad/s}$$

and hence the frequency ratio r is

$$r = \frac{\omega}{\omega_n} = \frac{5.81778}{18.2574} = 0.318653$$



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Solution (cont):

Note: $$\frac{X}{Y} = \left[\frac{k^2 + (c\omega)^2}{(k - m\omega^2)^2 + (c\omega)^2}\right]^{1/2} = \left[\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}\right]^{1/2} \quad (3.68)$$

The amplitude ratio can be found from Eq. (3.68):

$$\frac{X}{Y} = \left[\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}\right]^{1/2} = \left[\frac{1 + (2 \times 0.5 \times 0.318653)^2}{(1 - 0.318653)^2 + (2 \times 0.5 \times 0.318653)^2}\right]^{1/2}$$

$$= 1.100964$$

Thus the displacement amplitude of the vehicle is given by

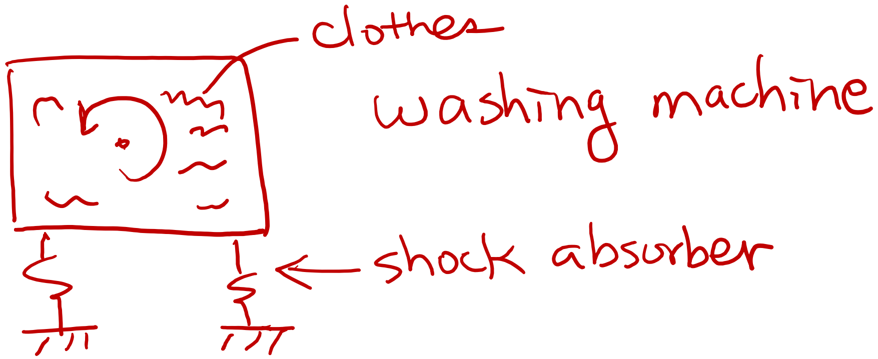
$$X = 1.100964Y = 1.100964(0.05) = 0.055048 \text{ m}$$

This indicates that a 5-cm bump in the road is transmitted as a 5.5-cm bump to the chassis and the passengers of the car. Thus in the present case, the passengers feel an amplified motion

Example – machine supported on foundation:

A heavy machine, weighing 3000 N, is supported on a resilient foundation. The static deflection of the foundation due to the weight of the machine is found to be 7.5 cm. It is observed that the machine vibrates with an amplitude of 1 cm when the base of the foundation is subjected to harmonic oscillation at the undamped natural frequency of the system with an amplitude of 0.25 cm. Find

a. the damping constant of the foundation, we don't know ζ
b. the dynamic force amplitude on the base, and
c. the amplitude of the displacement of the machine relative to the base.



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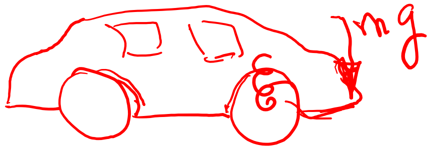
Solution:



vehicle suspension stiffness

$W_{person} = 100 \times 9.81 \simeq 1000\text{ N}, \quad \delta_{st} = 2\text{ cm}$

a. The stiffness of the foundation can be found from its static deflection: k = weight of machine$/\delta_{st} = 3000/0.075 = 40{,}000$ N/m.

$k \simeq \dfrac{1000}{0.02} = 50\text{ kN/m}$

At resonance ($\omega = \omega_n$ or $r = 1$), Eq. (3.68) gives

$$\frac{X}{Y} = \frac{0.010}{0.0025} = 4 = \left[\frac{1 + (2\zeta)^2}{(2\zeta)^2}\right]^{1/2} \tag{E.1}$$

The solution of Eq. (E.1) gives $\zeta = 0.1291$. The damping constant is given by

$$c = \zeta \cdot c_c = \zeta 2\sqrt{km} = 0.1291 \times 2 \times \sqrt{40{,}000 \times (3000/9.81)}$$
$$= 903.0512 \text{ N-s/m} \tag{E.2}$$

Note: $$\frac{X}{Y} = \left[\frac{k^2 + (c\omega)^2}{(k - m\omega^2)^2 + (c\omega)^2}\right]^{1/2} = \left[\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}\right]^{1/2} \tag{3.68}$$

Solution (cont):

b. The dynamic force amplitude on the base at $r = 1$ can be found from Eq. (3.74):

$$F_T = kY\left[\frac{1 + 4\zeta^2}{4\zeta^2}\right]^{1/2} = kX = 40{,}000 \times 0.01 = 400 \text{ N} \qquad \text{(E.3)}$$

c. The amplitude of the relative displacement of the machine at $r = 1$ can be obtained from Eq. (3.77):

$$Z = \frac{Y}{2\zeta} = \frac{0.0025}{2 \times 0.1291} = 0.00968 \text{ m} \qquad \text{(E.4)}$$

It can be noticed that $X = 0.01$ m, $Y = 0.0025$ m, and $Z = 0.00968$ m; therefore, $Z \neq X - Y$. This is due to the phase differences between x, y, and z.

Note:

$$\frac{F_T}{kY} = r^2\left[\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}\right]^{1/2} \qquad (3.74)$$

$$Z = \frac{m\omega^2 Y}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}} = Y\frac{r^2}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} \qquad (3.77)$$

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Example:

Derive the transfer function of a viscously damped single-degree-of-freedom system subjected to external force $f(t)$ as shown in Fig. 3.1.

$G(s)$

– linear systems

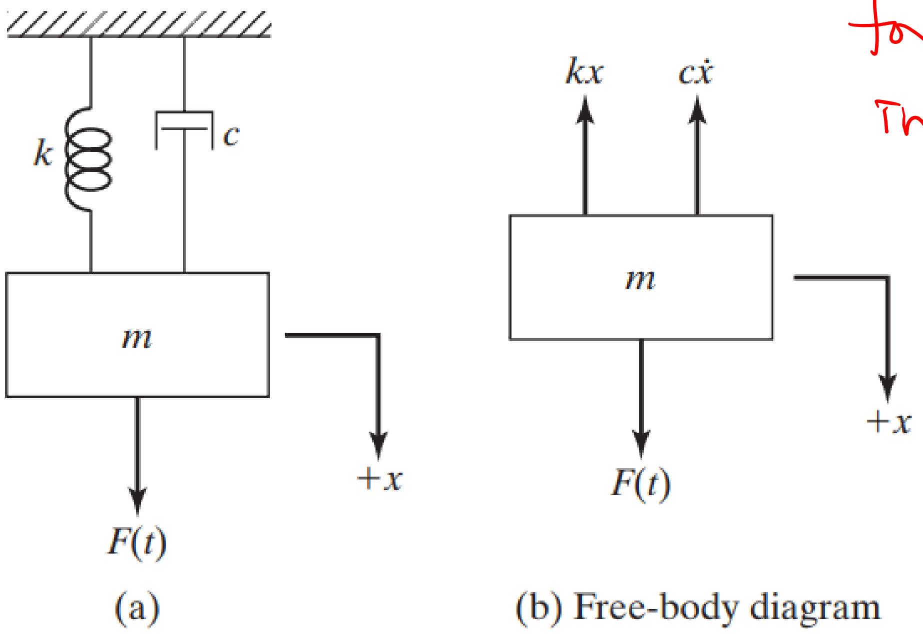


FIGURE 3.1 A spring-mass-damper system.

force

Input $\rightarrow \boxed{G(s)} \rightarrow$ output

response

s = frequency variable (Laplace transform)

$$\bar{X}(s) = \int_0^\infty e^{-st} x(t)\, dt$$

Solution: The equation of motion of the system is given by

$$m\ddot{x} + c\dot{x} + kx = f(t) \tag{E.1}$$

By taking the Laplace transforms of both sides of Eq. (E.1), we obtain

$$m\mathcal{L}[\ddot{x}(t)] + c\mathcal{L}[\dot{x}(t)] + k\mathcal{L}[x(t)] = \mathcal{L}[f(t)] \tag{E.2}$$

or

$$m\left[s^2X(s) - sx(0) - \dot{x}(0)\right] + \left[sX(s) - x(0)\right] + kX(s) = F(s) \tag{E.3}$$

Solution (cont):

Equation (E.3) can be rewritten as

$$(ms^2 + cs + k)\,\underline{X(s)} - \overbrace{[msx(0) + m\dot{x}(0) + sx(0)]}^{\text{gives free response}} = \overbrace{F(s)}^{\text{gives forced response}} \tag{E.4}$$

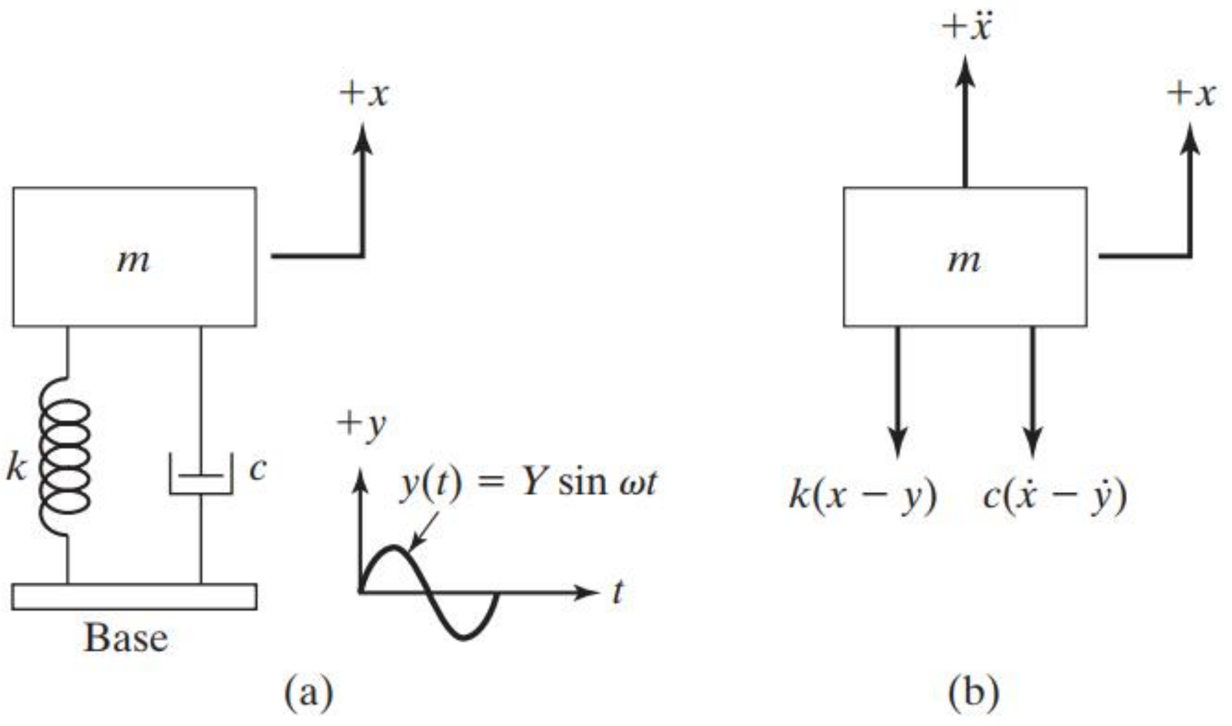
where $X(s) = \mathscr{L}[x(t)]$ and $F(s) = \mathscr{L}[f(t)]$. The transfer function of the system can be obtained from Eq. (E.4), by setting $x(0) = \dot{x}(0) = 0$, as

$$T(s) = \frac{\mathscr{L}[\text{output}]}{\mathscr{L}[\text{input}]}\bigg|_{\text{zero initial conditions}} = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k} \tag{E.5}$$

Transfer functions for linear time-invariant (LTI) systems can be also be derived using the general form of response due to harmonic excitation, assuming solution form of $X(\omega) = G(\omega)F(\omega)$.

Example – base excitation (numerical solution):

Using MATLAB, find and plot the response of a viscously damped spring-mass system under the base excitation $y(t) = Y \sin \omega t$ for the following data: $m = 1200$ kg, $k = 4 \times 10^5$ N/m, $\zeta = 0.5$, $Y = 0.05$ m, $\omega = 29.0887$ rad/s, $x_0 = 0$, $\dot{x}_0 = 0.1$ m/s.



Source: Rao, *Mechanical Vibrations*, 6e.

FIGURE 3.14 Base excitation.

Example – base excitation (numerical solution):

Solution: The equation of motion,

$$m\ddot{x} + c\dot{x} + kx = ky + c\dot{y} \tag{E.1}$$

can be expressed as a system of two first-order ordinary differential equations (using $x_1 = x$ and $x_2 = \dot{x}$) as

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{c}{m}x_2 - \frac{k}{m}x_1 + \frac{k}{m}y + \frac{c}{m}\dot{y} \tag{E.2}$$

with $c = \zeta c_c = 2\zeta\sqrt{km} = 2(0.5)\sqrt{(4 \times 10^5)(1200)}$, $y = 0.5 \sin 29.0887t$, and $\dot{y} = (29.0887)(0.05) \cos 29.0887t$. The MATLAB solution of Eq. (E.2), using `ode23`, is given below.

Solution (cont):

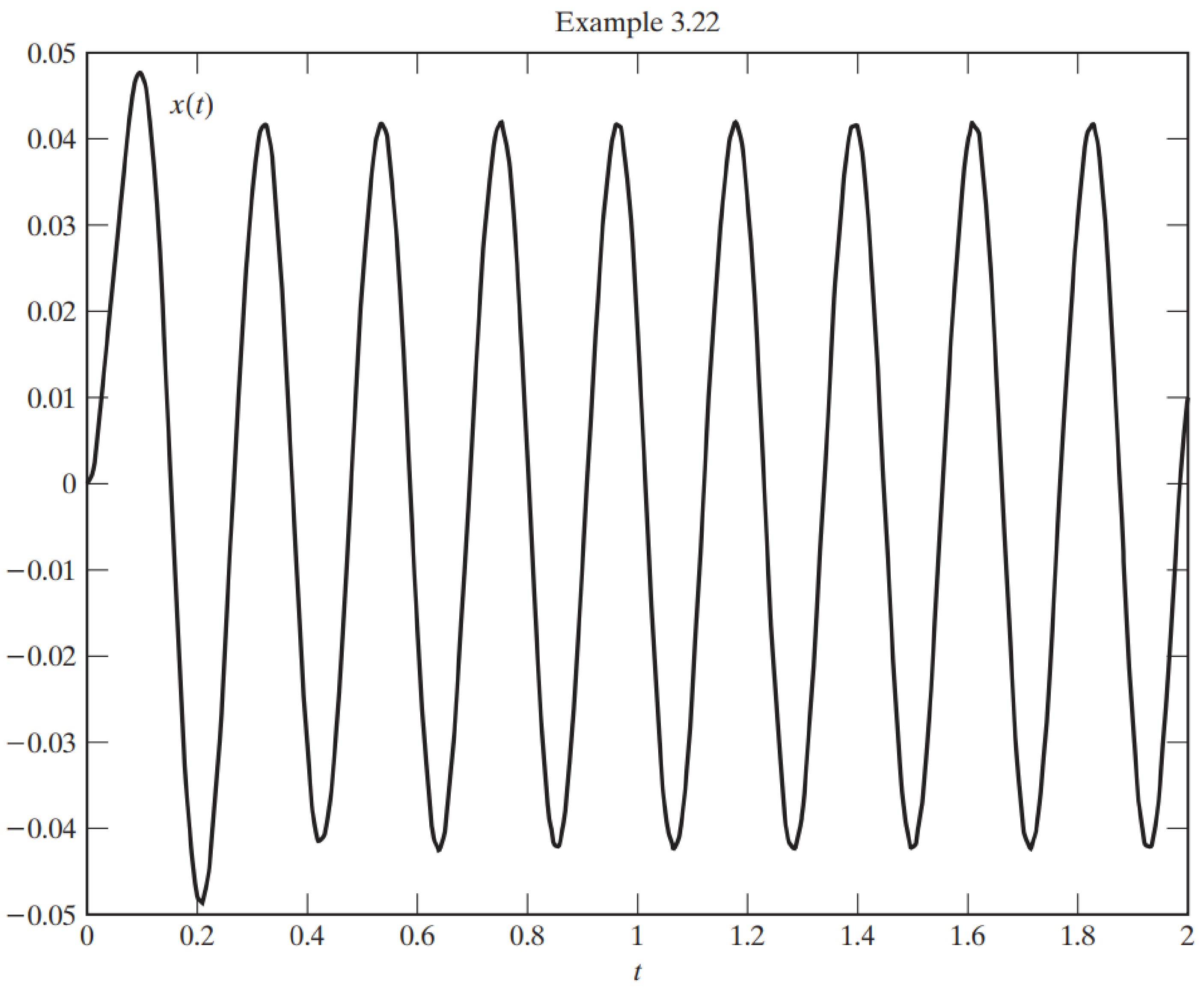
```
% Ex3_22.m
% This program will use the function dfunc3_22.m, they should
% be in the same folder
tspan = [0: 0.01: 2];
x0 = [0; 0.1];
[t, x] = ode23 ('dfunc3_22', tspan, x0);
disp ('      t           x(t)      xd(t)');
disp ([t x]);
plot (t, x (:, 1));
xlabel ('t');
gtext ('x(t)');
title ('Ex3.22');

% dfunc3_22.m
function f = dfunc3_22 (t, x)
f = zeros (2, 1);
f(1) = x(2);
f(2) = 400000*0.05*sin(29.0887*t)/1200 + ...
    sqrt (400000*1200)*29.0887*0.05*cos(29.0887*t)/1200 ...
    - sqrt(400000*1200)*x(2)/1200 - (400000/1200)*x(1);
```

Solution (cont):

Result:

```
>> Ex3_22
   t              x(t)         xd(t)
      0               0       0.1000
 0.0100          0.0022       0.3422
 0.0200          0.0067       0.5553
 0.0300          0.0131       0.7138
 0.0400          0.0208       0.7984
 0.0500          0.0288       0.7976
 .
 .
 .
 1.9500         -0.0388       0.4997
 1.9600         -0.0322       0.8026
 1.9700         -0.0230       1.0380
 1.9800         -0.0118       1.1862
 1.9900          0.0004       1.2348
 2.0000          0.0126       1.1796
```



3.1.2 Response to harmonic excitations (transfer function concept)

$$m\ddot{x}(t) + c\dot{x} + kx = F(t) = kA\cos\omega t \Rightarrow \ddot{x}(t) + 2\zeta\omega_n\dot{x} + \omega_n^2 x = A\omega_n^2\cos\omega t$$

$$x(t) = X\cos(\omega t - \phi) \quad \Rightarrow \quad \frac{X}{A} = \underbrace{\frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}}_{\text{magnification factor } M} = |G(i\omega)|\,, \qquad \phi = \tan\left(\frac{2\zeta r}{1-r^2}\right)$$

Complex notations: $F(t) = kA\,e^{i\omega t}, x(t) = X(\omega)\,e^{i\omega t},\ X(\omega) = A\,G(\omega)$, where

$$\boxed{G(\omega) = \frac{1}{(1-r^2) + 2i\zeta r}, \qquad \text{where } r = \omega/\omega_n}$$

Notes:

1) G(ω) is the frequency response function (FRF) or the system transfer function.
2) G(ω) defines the input (force) and output (response) relationship.
3) Response amplitude: X = A|G(ω)|. G(ω) also contains the phase information.
4) The poles (singularities) of G(ω) are the eigenvalues of the system. Physically, these correspond to the resonance or the peak response amplitudes.

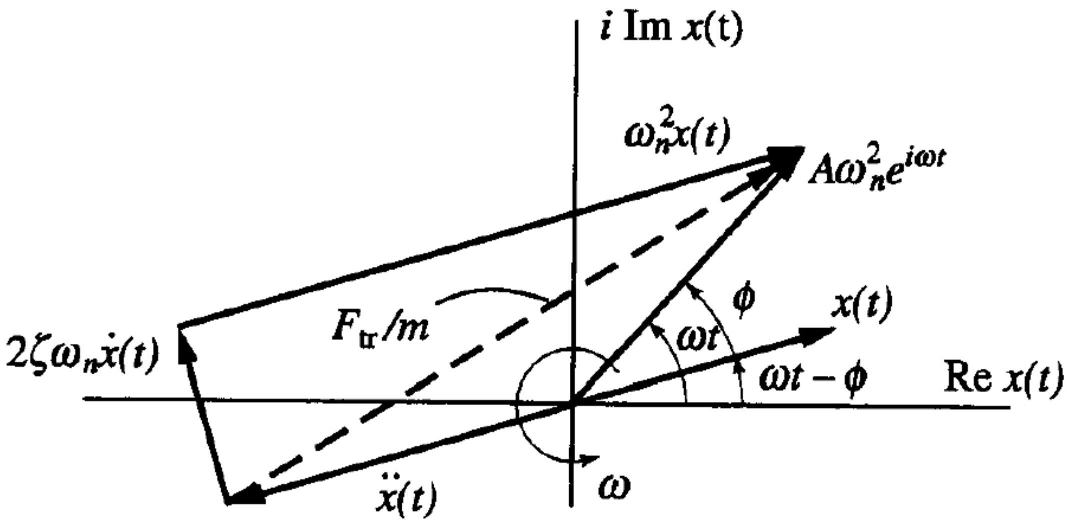


Fig. 2.5 Illustration of phase difference.

3.2 Damping Measurements under harmonic excitation

- Unlike the measurement of the spring constant by static tests (tensile tests), measurement of the damping constant requires dynamic tests.
- The logarithmic decrement method discussed before is a time domain method based on the free vibration. <u>Difficulty of this method is that it does not provide a measurement approach to obtain multiple damping ratios</u>.
- The half-power method introduced here is based on the steady-state frequency response under harmonic excitation.

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Consider the equation of motion: $\ddot{x}+2\zeta\omega_n\dot{x}+\omega_n^2 x = ae^{j\omega t}$ (3.19)

In passing, note here that the frequency-domain transfer function (i.e., response/excitation in the frequency domain) of the system (3.19) is:

$$G(j\omega)=\frac{1}{\Delta}=\left[\frac{1}{s^2+2\zeta\omega_n s+\omega_n^2}\right]_{s=j\omega} \tag{3.34}$$

Equation (3.28c) is: $\Delta=\omega_n^2-\omega^2+2\zeta\omega_n\omega j$
Hence,

$$|\Delta|^2=\left(\omega_n^2-\omega^2\right)^2+\left(2\zeta\omega_n\omega\right)^2=D \tag{3.35}$$

The resonance corresponds to a minimum value of D; or maximum of |G|

$$\frac{dD}{d\omega}=2\left(\omega_n^2-\omega^2\right)(-2\omega)+2\left(2\zeta\omega_n\right)^2\omega=0 \qquad \text{(for a minimum)} \tag{3.36}$$

This is the *resonant frequency*, and is denoted as

$$\omega_r=\sqrt{1-2\zeta^2}\;\omega_n \tag{3.37}$$

Note the difference between this frequency and the damped natural frequency

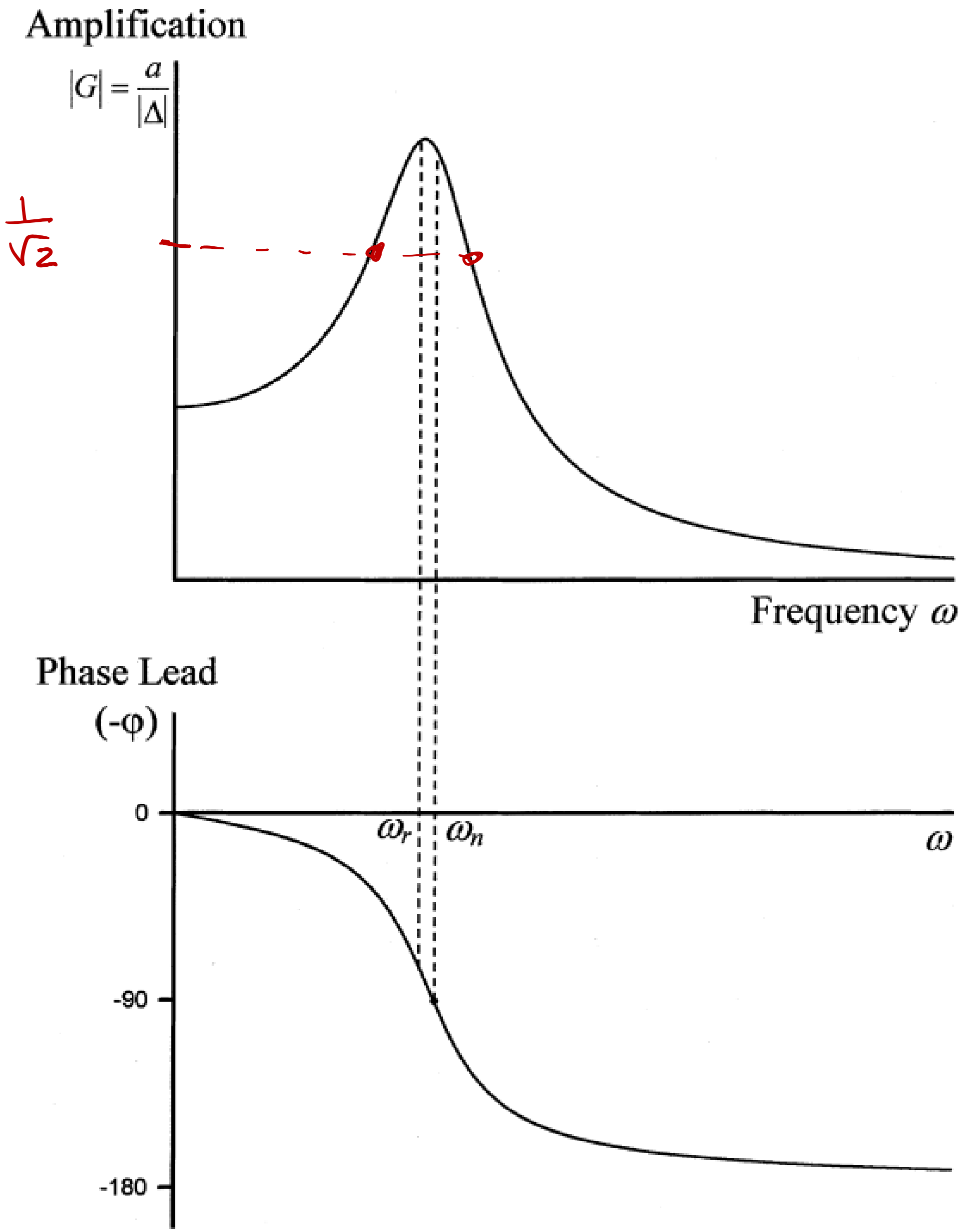


FIGURE 3.3 Magnitude and phase angle curves of a simple oscillator (a Bode plot).

3.1.2 Measurement of Damping Ratio (Q-Factor Method)

The frequency transfer function of a simple oscillator (3.19) can be used to determine the damping ratio. This frequency-domain method is also termed the *half-power point* method, for reasons that should be clear from the following development.

First assume that $\zeta < \frac{1}{\sqrt{2}}$. Strictly speaking, one should assume that $\zeta < \frac{1}{2\sqrt{2}}$.

Without loss of generality, consider the normalized (or nondimensionalized) transfer function

$$G(j\omega) = \left[\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}\right]_{s=j\omega} = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + 2\zeta\omega_n\omega j} \tag{3.38}$$

As noted before, the transfer function $G(s)$, where s is the Laplace variable, can be converted into the corresponding frequency transfer function by simply setting $s = j\omega$. Its value at the undamped natural frequency is

$$G(j\omega)\big|_{\omega=\omega_n} = \frac{1}{2\zeta j} \tag{3.39a}$$

Hence, the magnitude of $G(j\omega)$ (amplification) at $\omega = \omega_n$ is

$$\left|G(j\omega)\right|_{\omega=\omega_n} = \frac{1}{2\zeta} \tag{3.39b}$$

For small ζ, $\omega_r \cong \omega_n$. Hence, $\frac{1}{2\zeta}$ is approximately the peak magnitude at resonance (resonant peak). The actual peak is slightly larger.

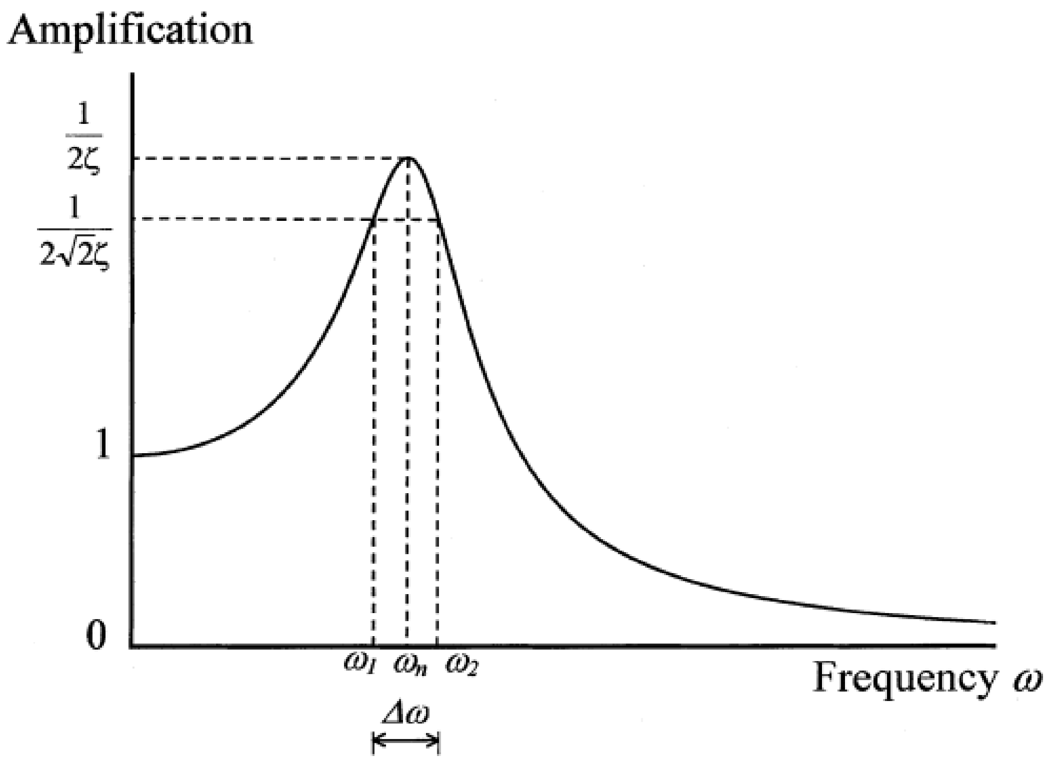


FIGURE 3.4 The Q-factor method of damping measurement.

Hence, the damping ratio

$$\zeta \cong \frac{(\omega_2 - \omega_1)}{2\omega_n} = \frac{\Delta\omega}{2\omega_n} \cong \frac{\omega_2 - \omega_1}{\omega_2 + \omega_1} \tag{3.44}$$

It follows that, once the magnitude of the frequency response function $G(j\omega)$ is experimentally determined, the damping ratio can be estimated from equation (3.44), as illustrated in Figure 3.4. The Q-factor, which measures the sharpness of resonant peak, is defined by

$$\text{Q-factor} = \frac{\omega_n}{\Delta\omega} = \frac{1}{2\zeta} \tag{3.45}$$

Example 3.1

A dynamic model of a fluid coupling system is shown in Figure 3.5. The fluid coupler is represented by a rotatory viscous damper with damping constant b. It is connected to a rotatory load of moment of inertia J, restrained by a torsional spring of stiffness k, as shown. Obtain the frequency transfer function of the system relating the restraining torque τ of the spring to the angular displacement excitation $\alpha(t)$ that is applied at the free end of the fluid coupler. If $\alpha(t) = \alpha_o \sin\omega t$, what is the magnitude (i.e., amplitude) of τ at steady state?

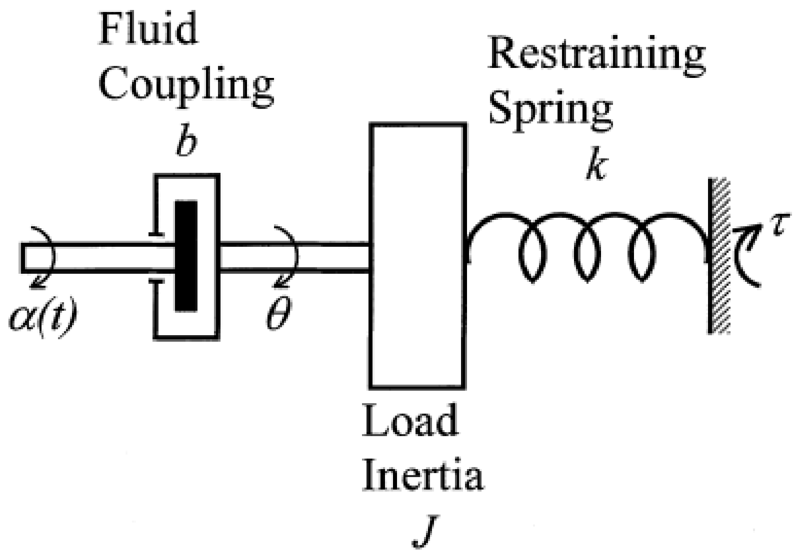


FIGURE 3.5 A fluid coupling system.

Governing equation of motion:

$$J\ddot{\theta} = b\left(\dot{\alpha} - \dot{\theta}\right) - k\theta$$

Motion transfer function is

$$\frac{\theta}{\alpha} = \frac{bs}{Js^2 + bs + k}$$

Restraining torque of the spring is $\tau = k\theta$. Hence,

$$\frac{\tau}{\alpha} = \frac{k\theta}{\alpha} = \frac{kbs}{Js^2 + bs + k} = G(s)$$

Then, the corresponding frequency response function (frequency transfer function) is

$$G(j\omega) = \frac{kbj\omega}{\left(k - J\omega^2\right) + bj\omega}$$

For a harmonic excitation of

$$\alpha = \alpha_o e^{j\omega t}$$

one has

$$\tau = \left|G(j\omega)\right| \alpha_o e^{(j\omega t + \phi)}$$

at steady state.
Here, the phase "lead" of τ with respect to α is

$$\begin{aligned} \phi &= \angle G(j\omega) \\ &= \angle j\omega - \angle\left(k - J\omega^2 + bj\omega\right) \\ &= \frac{\pi}{2} - \tan^{-1} \frac{b\omega}{\left(k - J\omega^2\right)} \end{aligned}$$

The magnitude of the restraining torque at steady state is

$$\begin{aligned} \tau_o &= \alpha_o \left|G(j\omega)\right| \\ &= \alpha_o \frac{kb\omega}{\left|k - J\omega^2 + bj\omega\right|} = \frac{\alpha_o kb\omega}{\sqrt{\left(k - J\omega^2\right)^2 + b^2\omega^2}} \end{aligned}$$

For $r = 1$: $\tau_o = \dfrac{2k\alpha_o\zeta}{2\zeta} = k\alpha_o$ (*xii*)

This means, at resonance, the applied twist is directly transmitted to the load spring.

For small r: $\tau_o = \dfrac{2k\alpha_o\zeta r}{1}$ (*xiii*)

which is small, and becomes zero at $r = 0$. Hence, at low frequencies, the transmitted torque is small.

For large r: $\tau_o = \dfrac{2k\alpha_o\zeta r}{r^2} = \dfrac{2k\alpha_o\zeta}{r}$ (*xiv*)

which is small, and goes to zero. Hence, at high frequencies as well, the transmitted torque is small.

The variation of τ_o with the frequency ratio r is sketched in Figure 3.6.

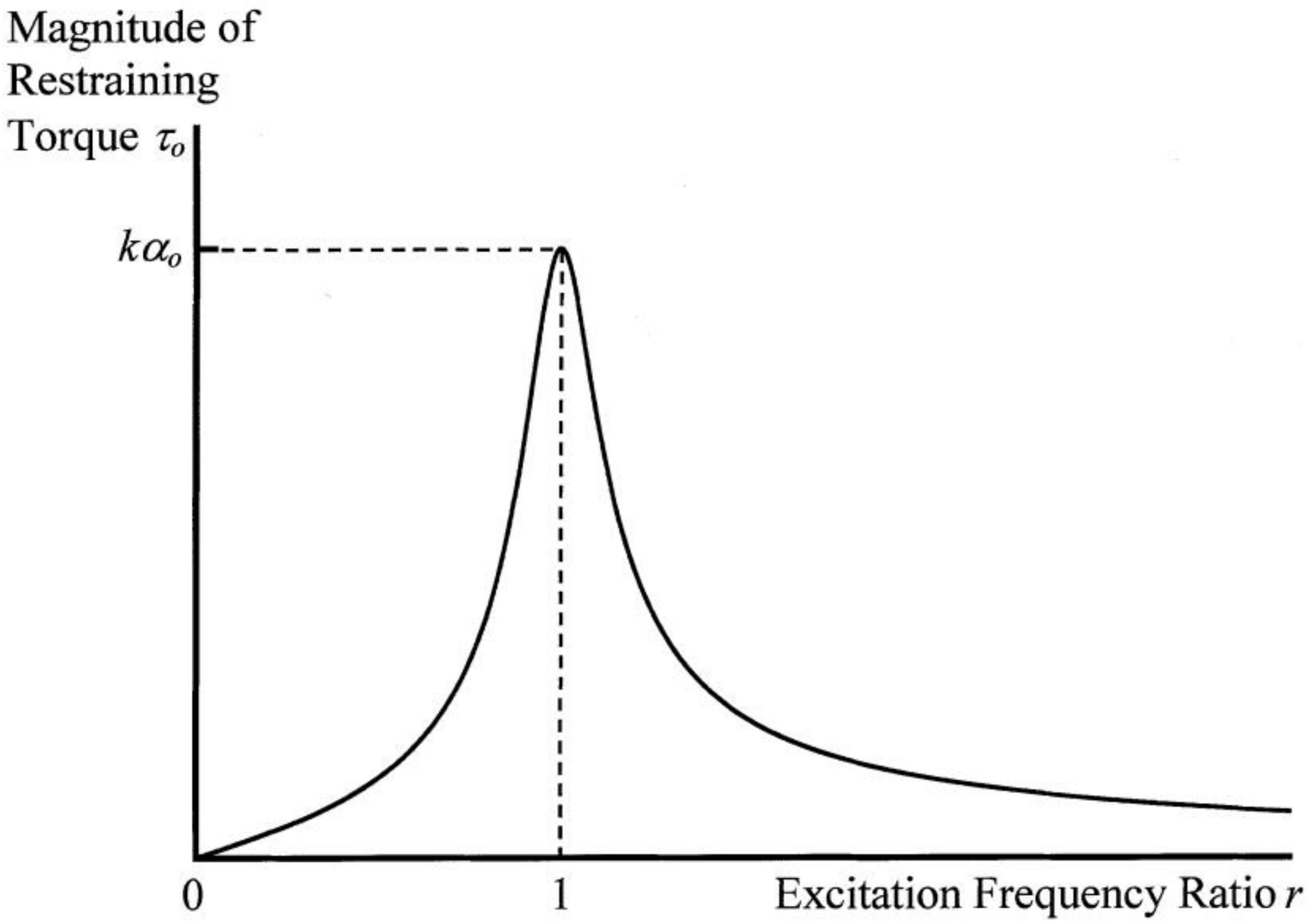
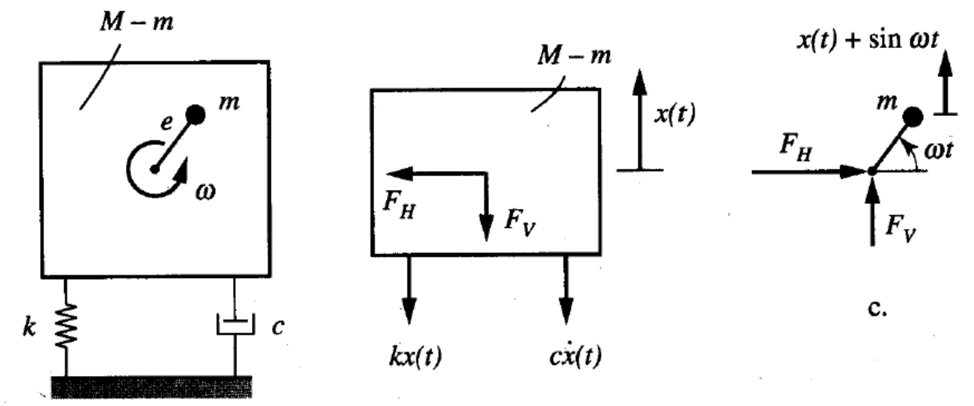


FIGURE 3.6 Variation of the steady-state transmitted torque with frequency.

2.2.1.1 Example: Systems with rotating unbalanced masses

$$(M-m)\ddot{x} = -kx - c\dot{x} - F_V\ , \quad F_V = \frac{d^2}{dt^2}\left(x + e\sin\omega t\right)$$

$$\ddot{x}(t) + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \underbrace{\frac{me}{M}\omega^2}_{"A\omega_n^2"}\cos\omega t\ , \quad x(t) = \frac{me}{M}\left(\frac{\omega}{\omega_n}\right)^2 \left|G(i\omega)\right| \sin(\omega t - \phi)$$



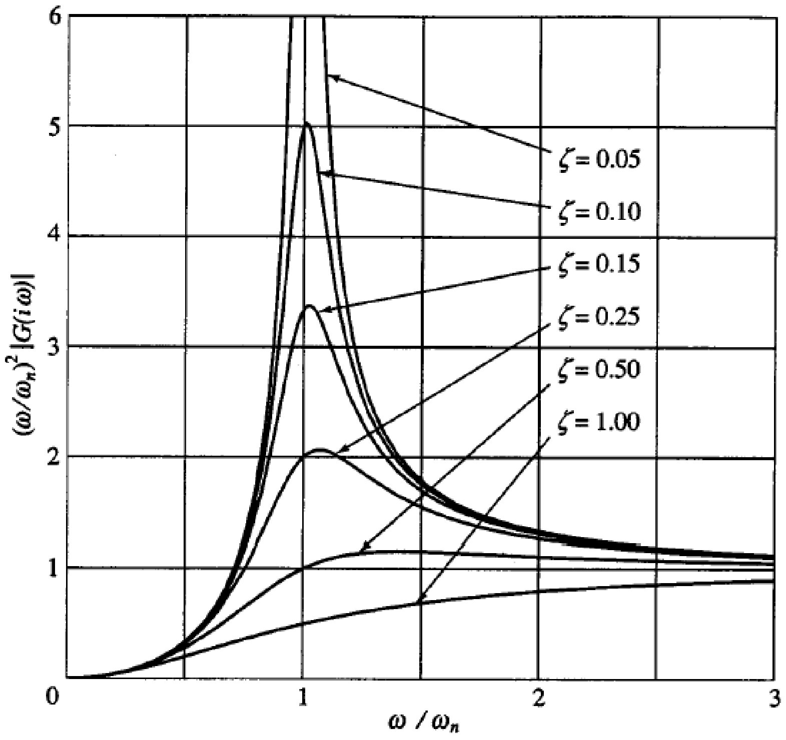
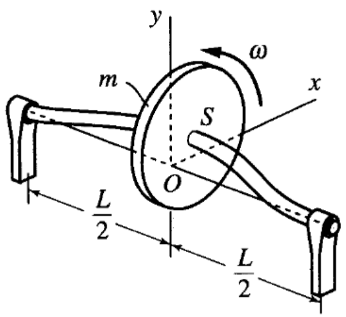


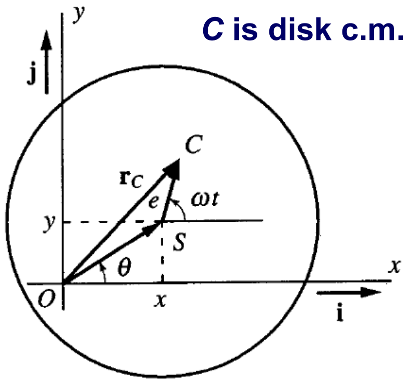
Fig. 2.6 Frequency response magnitude.

Notes:

1) Obtain the system response by the frequency response function
2) No static response
3) Peak amplitude occurs at $\omega > \omega_n$
4) The frequency response plot approaches 1 for large ω
5) The phase angle approaches π for large ω

2.2.1.2 Example: Whirling of rotating shafts with eccentric disks





C is disk c.m.

$$\ddot{x}(t)+2\zeta_x\omega_{nx}\dot{x}+\omega_{nx}^2x=e\omega^2\cos\omega t\ ,$$
$$\ddot{y}(t)+2\zeta_y\omega_{ny}\dot{y}+\omega_{ny}^2y=e\omega^2\sin\omega t$$

Note: Each equation is in the "standard" form and represents a system with unbalance rotating mass.

$$x(t)=e\left(\frac{\omega}{\omega_{nx}}\right)^2\left|G_x(i\omega)\right|\cos(\omega t-\phi_x),\ y(t)=e\left(\frac{\omega}{\omega_{ny}}\right)^2\left|G_y(i\omega)\right|\sin(\omega t-\phi_y)$$

Circular cross-section case:

$$\omega_{nx}=\omega_{ny}=\omega,\ \zeta_x=\zeta_y=\zeta\ ,\ \phi_x=\phi_y=\phi$$

$$\tan\theta=y/x=\tan(\omega t-\phi),\ \theta=\omega t-\phi,\ \dot{\theta}=\omega\quad(\dot{\theta}=\text{shaft angular speed})$$

Synchronous Whirl:

1) **Shaft whirls with the same angular velocity as the disk, so that the shaft and the disk rotate together as a rigid body.**
2) **Distance from *O* to *S* for a given ω is constant.**

Asynchronous whirling (no damping) case:

$$x(t) = X(\omega)\cos\omega t\ , \quad y(t) = Y(\omega)\sin\omega t$$

$$X(\omega) = \frac{e(\omega/\omega_{nx})^2}{1-(\omega/\omega_{nx})^2}\ , \quad Y(\omega) = \frac{e(\omega/\omega_{ny})^2}{1-(\omega/\omega_{ny})^2} \qquad \frac{x^2}{X^2(\omega)} + \frac{y^2}{Y^2(\omega)} = 1\ , \quad \tan\theta = \frac{Y}{X}\tan\omega t$$

- **Differentiate above expression and simplify**

$$\dot{\theta} = \frac{X\,Y}{X^2(\omega)\cos^2\omega t + Y^2(\omega)\sin^2\omega t}\,\omega$$

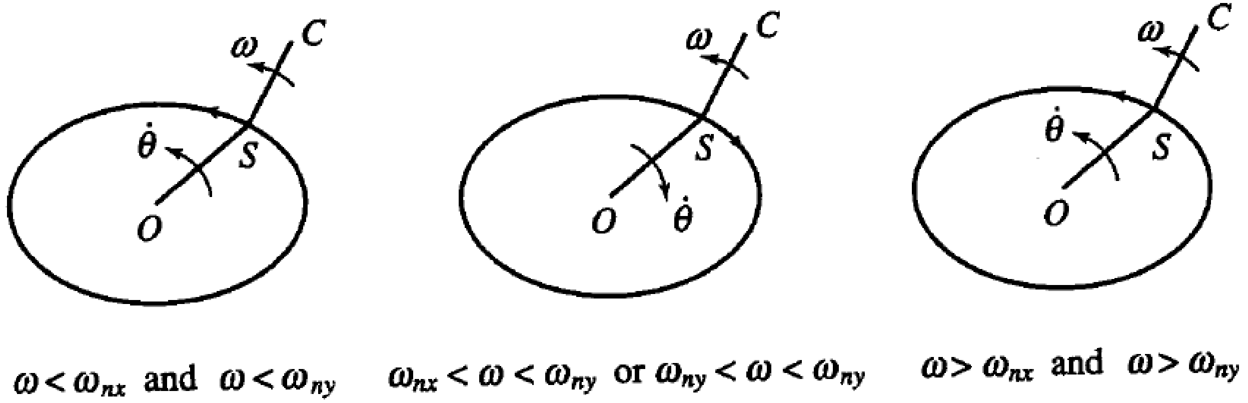
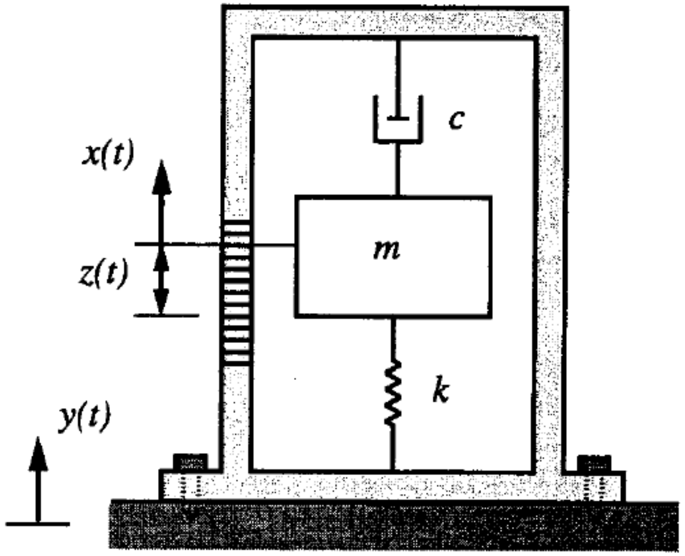


Fig. 2.7 Asynchronous rotations of shaft and rotor.

Case 1: $\omega < \omega_{nx}$ and $\omega < \omega_{ny}$. XY > 0, so S moves on an ellipse in the same sense as ω.
Case 2: $\omega_{nx} < \omega < \omega_{ny}$ or $\omega_{ny} < \omega < \omega_{nx}$. XY < 0, so S moves in the opposite sense.
Case 3: $\omega > \omega_{nx}$ and $\omega > \omega_{ny}$. XY > 0, so S moves in the same sense.

We can view ω_{nx} and ω_{ny} as critical speeds.

2.2.1.3 Example: Vibration measuring instrument



absolute displacement of the mass:

$$x(t) = y(t) + z(t)$$

displacement/excitation of device:

$$y(t) = Y_0 \cos \omega t$$

equation of motion for the mass:

$$m\ddot{z}(t) + c\dot{z} + kz = Y_0 m\omega^2 \cos \omega t$$

$$\frac{Z_0}{Y_0} = \left(\frac{\omega}{\omega_n}\right)^2 |G(i\omega)|$$

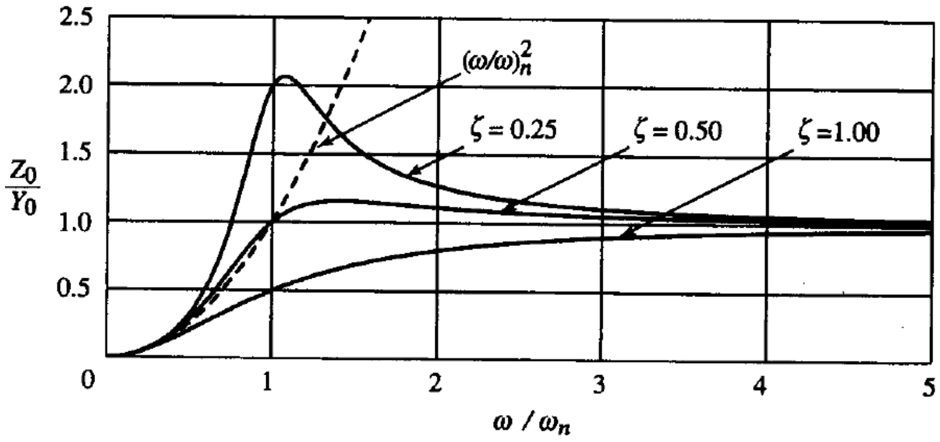


Fig. 2.8 Frequency response magnitude for accelerometer devices.

Notes:

1) Objective is to determine the motion *y*(*t*) from measurements of *z*(*t*).
2) For $|G(i\omega)|$ almost unity, Z_0 is proportional to the device acceleration $Y_0\omega^2$. This is how an accelerator works.
3) Typically, ω_n ~ 1-5 Hz, ω ~ 10-500 Hz.
4) Application to seismometers.

2.2.2 Concept of structural damping:

- In addition to viscous and Coulomb damping, there is also energy dissipation due to internal friction among molecules. This is called structural damping, attributed to *hysteresis* phenomenon associated with cyclic stress in elastic materials.

- Energy loss in hysteresis:

$$\Delta E_{\text{cycle}} = \alpha X^2$$

- Energy loss in viscous damping:

$$\Delta E_{\text{cycle}} = c\pi\omega X^2$$

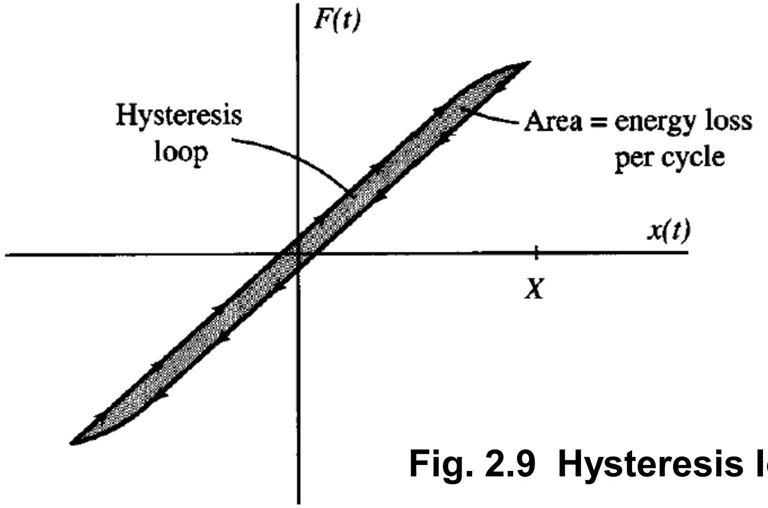
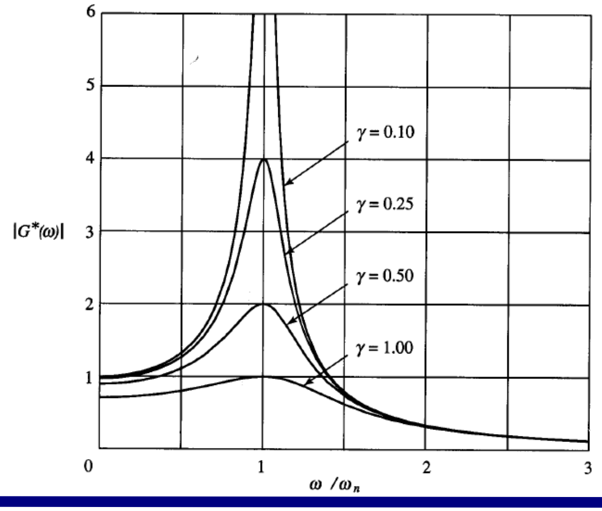


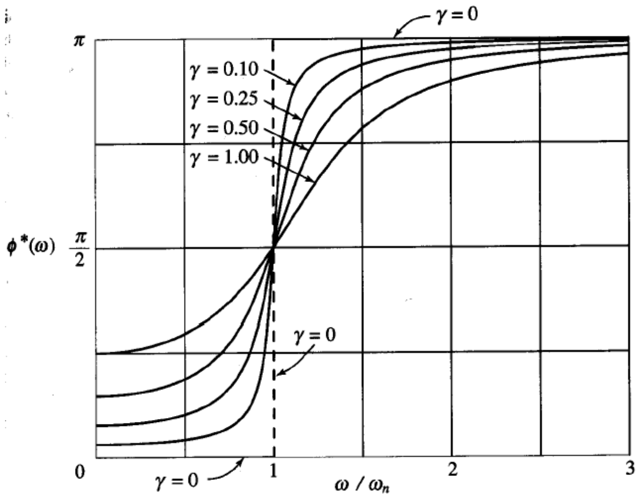
Fig. 2.9 Hysteresis loop.

$$m\ddot{x}(t) + k(1+i\gamma)x = kAe^{i\omega t}, \quad \gamma = \frac{\alpha}{\pi k},$$

$$G^*(\omega) = \frac{1}{1-(\omega/\omega_n)^2 + i\gamma}$$

Fig. 2.10 Magnitude and phase of frequency response.





2.2.3 Response to periodic excitations:

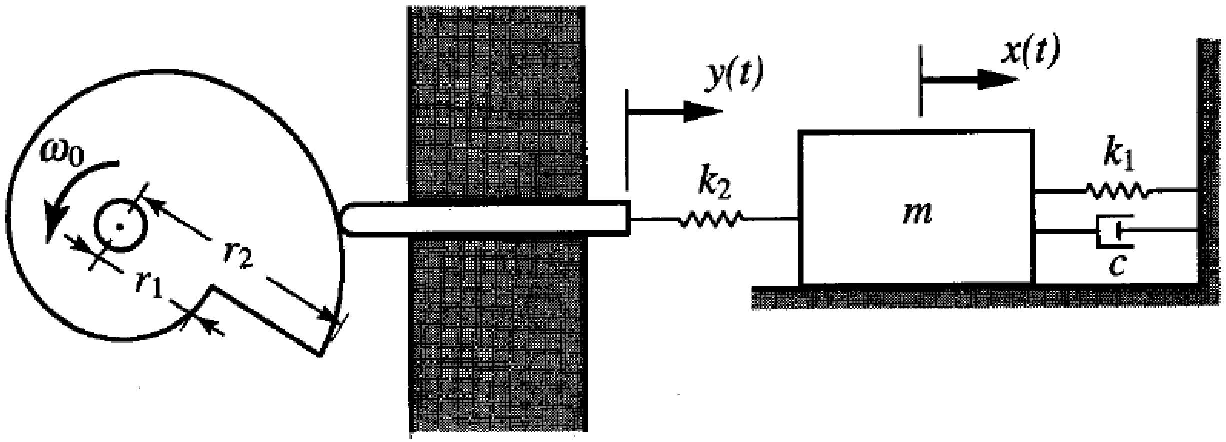
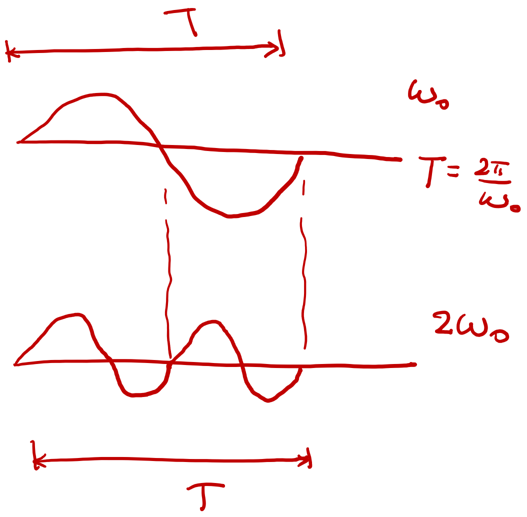


Fig. 2.11 Cam and follower exerting a periodic force.



- **Excitation in a Fourier series => application of superposition principle**

$$f(t)=\frac{a_0}{2}+\sum_{p=1}^{\infty}\underbrace{(a_p\cos p\omega_0 t+b_p\sin p\omega_0 t)}_{\text{harmonics}}\,,\qquad \omega_0=2\pi/T$$

one fundamental frequency

Fourier coefficients a_p, b_p

- **Fourier series (Fourier transform for general signals)**

$$a_p=\frac{2}{T}\int_0^T f(t)\cos p\omega_0 t\,dt\,,\quad b_p=\frac{2}{T}\int_0^T f(t)\sin p\omega_0 t\,dt$$

The concept of Fourier series is based on the expansion or decomposition of $f(t)$ in terms of a set of **orthogonal basis functions**. This basic concept of expansion was introduced in elementary vector calculus. Consider a 3D vector **a** written in terms of basis vectors:

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3$$

The extraction of the components is given by:

$$a_j = (\mathbf{a}\cdot\mathbf{e}_j)/(\mathbf{e}_j\cdot\mathbf{e}_j)$$

where the term in the denominator of the above expression is a normalization. Usually, the basis vectors are of unit magnitude. Hence $(\mathbf{e}_j\cdot\mathbf{e}_j) = 1$ and $a_j = (\mathbf{a}\cdot\mathbf{e}_j)$, i.e., this is the projection of the vector **a** along the j-th unit basis vector. Employing this concept, we express the periodic function $f(t)$ as:

$$f(t) = \frac{a_0}{2} + \sum_{p=1}^{\infty}\left(a_p \cos p\omega_0 t + b_p \sin p\omega_0 t\right)$$

where the sine and cosine functions are orthogonal basis functions over $[0, T]$ and are assumed to form a complete basis for any periodic function. To obtain the coefficients, employ the same concept as the dot product for vectors.

$$a_p = (f,\phi_p)/(\phi_p,\phi_p) \quad \text{where, } \phi_p = \cos p\omega_0 t$$

$$(f,\phi_p) = \int_0^T f(t)\cos p\omega_0 t \; dt\,, (\phi_p,\phi_p) = T/2$$

This decomposition concept is fundamental for the development of the modal analysis method in vibration.

- **Complex notations**

$$f(t)=\sum_{-\infty}^{\infty} C_p e^{ip\omega_0 t} = C_0 + \sum_{p=1}^{\infty} C_p e^{ip\omega_0 t} + \sum_{p=1}^{\infty} C_{-p} e^{ip\omega_0 t}$$

$$C_p = \frac{1}{T}\int_0^T f(t)\, e^{-ip\omega_0 t}\, dt\,, \quad p\in(-\infty,\infty)$$

***f*(*t*) is a real-valued function =>** $C_{-p}=C_p^*$ **(* denotes complex conjugate)**

- **Response by superposition principle; similar to the steady-state response obtained for harmonic excitations**

$$m\ddot{x}(t)+c\dot{x}+kx=kf(t)\,, \quad f(t)=\frac{A_0}{2}+\mathrm{Re}\left[\sum_{p=1}^{\infty} A_p e^{ip\omega_0 t}\right]$$

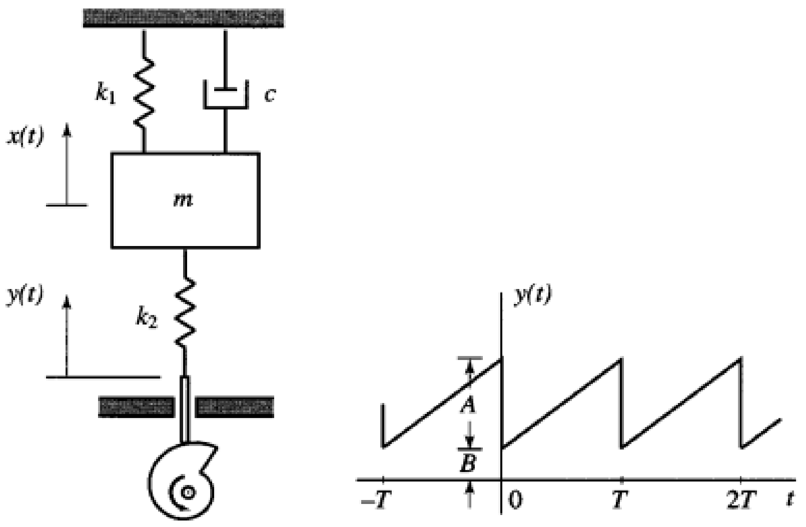
$$x(t)=\frac{A_0}{2}+\mathrm{Re}\left[\sum_{p=1}^{\infty} A_p \left|G_p\right| e^{i(p\omega_0 t-\phi_p)}\right], \qquad A_p=\frac{2}{T}\int_0^T f(t)\,e^{-ip\omega_0 t}\, dt,\, p=0,1,2,3,\ldots$$

$$G_p=\frac{1}{1-\left(p\omega_0/\omega_n\right)^2+i\,2\zeta\, p\omega_0/\omega_n}\,, \quad \phi_p=\tan^{-1}\left[\frac{2\zeta\, p\omega_0/\omega_n}{1-\left(p\omega_0/\omega_n\right)^2}\right]$$

8. [Meirovitch 3.18] The cam and follower impart a displacement $y(t)$ in the form of a periodic sawtooth function to the lower end of a vibratory system as shown in the figure.

a) Draw a free body diagram and derive the governing equation of motion.

b) Obtain the steady-state response $x(t)$ by means of a Fourier analysis.



c) Examine the *accuracy* of the series expression by evaluating the maximum-norm amplitude of the response summing over 2, 5, 10 and 20 terms. Let $A = 20$ cm, $B = 10$ cm, $T = 1$ sec., $m = 100$ kg, $c = 50$ N·s/m, and $k_1 = k_2 = 100$ N/m. Besides depending on the number of terms in the series, does the numerical accuracy of the series summation also depend on the system parameters? Please comment as much as you can.

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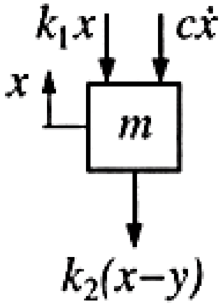
Free-body diagram and Fourier coefficients



$$m\ddot{x} = -c\dot{x} - k_1 x - k_2(x-y)$$

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \frac{k_2}{m}y$$

$$\text{where } 2\zeta\omega_n = \frac{c}{m},\ \omega_n^2 = \frac{k_1+k_2}{m}$$

$$\text{Over the period } 0 < t < T,\ y(t) = \frac{A}{T}t + B$$

$$y(t) = \frac{1}{2}A_0 + \mathrm{Re}\left(\sum_{p=1}^{\infty} A_p e^{ip\omega_0 t}\right)$$

$$A_0 = \frac{2}{T}\int_0^T y(t)dt = \frac{2}{T}\int_0^T \left(\frac{A}{T}t + B\right)dt = 2\left(\frac{A}{2} + B\right)$$

$$A_p = \frac{2}{T}\int_0^T y(t)e^{-ip\omega_0 t}dt = \frac{2}{T}\int_0^T \left(\frac{A}{T}t + B\right)e^{-ip\omega_0 t}dt = \frac{iA}{\pi p}$$

Response solution

The equation of motion becomes

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \frac{k_2}{m}\left[\frac{A}{2} + B + \frac{A}{\pi}\mathrm{Re}\left(\sum_{p=1}^{\infty}\frac{i}{p}e^{ip\omega_0 t}\right)\right]$$

The response is

$$\begin{aligned} x(t) &= \frac{k_2}{k_1+k_2}\left[\frac{A}{2} + B + \frac{A}{\pi}\mathrm{Re}\left(\sum_{p=1}^{\infty}\frac{i}{p}|G_p|e^{i(p\omega_0 t-\phi_p)}\right)\right] \\ &= \frac{k_2}{k_1+k_2}\left[\frac{A}{2} + B - \frac{A}{\pi}\sum_{p=1}^{\infty}\frac{1}{p}|G_p|\sin(p\omega_0 t - \phi_p)\right] \end{aligned}$$

where

$$|G_p| = \frac{1}{\{[1-(p\omega_0/\omega_n)^2]^2 + (2\zeta p\omega_0/\omega_n)^2\}^{1/2}}, \quad \phi_p = \tan^{-1}\frac{2\zeta p\omega_0/\omega_n}{1-(p\omega_0/\omega_n)^2}$$

Numerical accuracy of series solution to Meirovitch 3.18:

From Problem 3.18, the response solution was obtained as

$$x(t)=\frac{k_2}{k_1+k_2}\left[\frac{A}{2}+B-\frac{A}{\pi}\sum_{p=1}^{N\to\infty}\frac{1}{p}\left|G_p\right|\sin(p\omega_0 t-\phi_p)\right],$$

where $\omega_0=2\pi/T$. The magnitude and the phase of the frequency response function $G_p(\omega)$ are

$$\left|G_p\right|=\frac{1}{\left\{\left[1-(p\omega_0/\omega_n)^2\right]^2+(2\zeta p\omega_0/\omega_n)^2\right\}^{1/2}},\quad \phi_p=\tan^{-1}\left(\frac{2\zeta p\omega_0/\omega_n}{1-(p\omega_0/\omega_n)^2}\right).$$

We further define the max-norm of $x(t)$ as

$$\left\|x(t)\right\|_{\max}\triangleq\frac{k_2}{k_1+k_2}\left[\frac{A}{2}+B-\frac{A}{\pi}\sum_{p=1}^{N\to\infty}\frac{1}{p}\left|G_p\right|\right].$$

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For convergence study, we further define a percentage error as

$$\varepsilon_N \triangleq \frac{\left\|x_N(t)\right\|_{\max} - \left\|x_{20}(t)\right\|_{\max}}{\left\|x_{20}(t)\right\|_{\max}} \times 100\%$$

where $\left\|x_N(t)\right\|_{\max}$ denotes the max-norm with N terms in the series. Note that the relative error is scaled against the $\left\|x_{20}(t)\right\|_{\max}$ (N = 20 terms). A MATLAB program (next page) is written to plot the time history and calculate the max-norm as a function of the number of terms employed in the series. From Table 1, it can be seen that the convergence is excellent.

Table 1: Convergence of the maximum norm.

N	$\left\|x(t)\right\|_{\max}$	Percentage error, ε_N
1	0.09830730608256	0.51
2	0.09810331388038	0.30
5	0.09800507540602	0.20
10	0.09798591537784	0.19
20	0.09780534597987	0.00

Note that, since $\left|G_p\right| \sim \dfrac{1}{p}$, hence the max-norm $\left\|x(t)\right\|_{\max}$ converges as $1/p^2$ (quadratic convergence). In general, the accuracy also depends on the system parameters. For example, the parameter "A" serves as an error amplification factor in the series.

MATLAB Program:

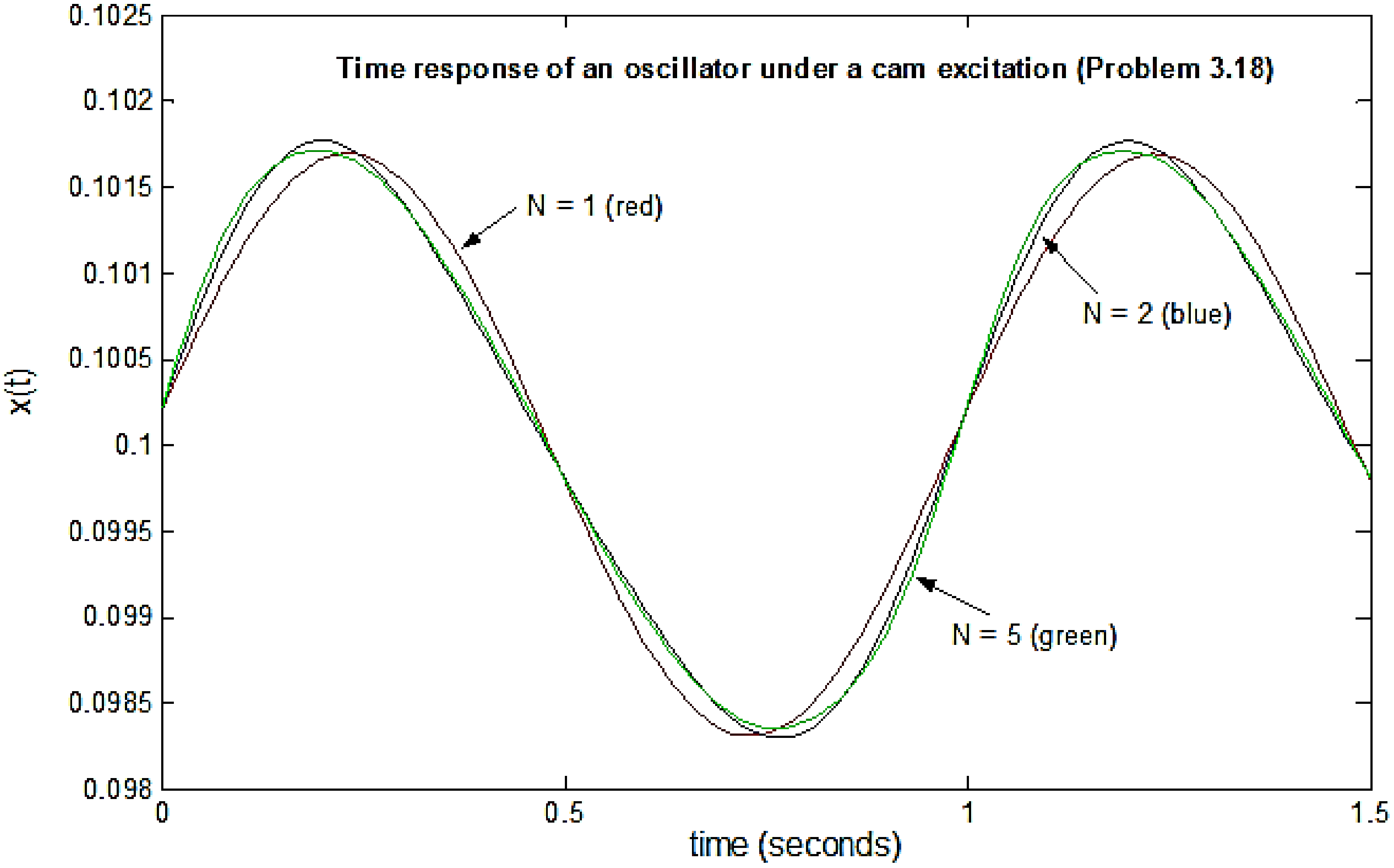
```
M=101;  % number of data points in t-domain; 100 intervals in [0, 1.5]
t=linspace(0,1.5,M);
% Define values of the physical parameters
m=100;
k1=100; k2 =100;
A =0.2;
B=0.1;
c=50;
T =1;
N = [1,2,5];
omega_n = sqrt( (k1+k2)/m);
omega_0 = 2*pi/T;
zeta = c/(2*m*omega_n);

for i=1:length(N)
    G = 0;
    phi = 0;
    x(i,:)= (k1/(k1+k2) *(A/2 + B)) * ones (1,M);
    xmax(i) = k1/(k1+k2) *(A/2 + B);

    for p=1:N(i)

    G = 1/sqrt((1-(p*omega_0/omega_n)^2)^2+(2*zeta*p*omega_0/omega_n)^2);
    phi= atan2( 2*zeta*p*omega_0, 1-(p*omega_0/omega_n)^2);
    x(i,:) = x(i,:) - k1/(k1+k2) * (A/pi/p) * abs(G)* sin(p*omega_0*t-phi);
    xmax(i) = xmax(i) - k1/(k1+k2) * (A/pi/p) * abs(G);
    end
    format long
    xmax(i)

end
plot(t, x(1,:),'r', t, x(2,:),'b', t, x(3,:),'g')
```


Time response of an oscillator under a cam excitation (Problem 3.18)
N = 1 (red)
N = 2 (blue)
N = 5 (green)
x(t)
time (seconds)
0.1025
0.102
0.1015
0.101
0.1005
0.1
0.0995
0.099
0.0985
0.098
0
0.5
1
1.5